

模拟电子技术基础

Analog Device

7 Signal operation

信号的运算

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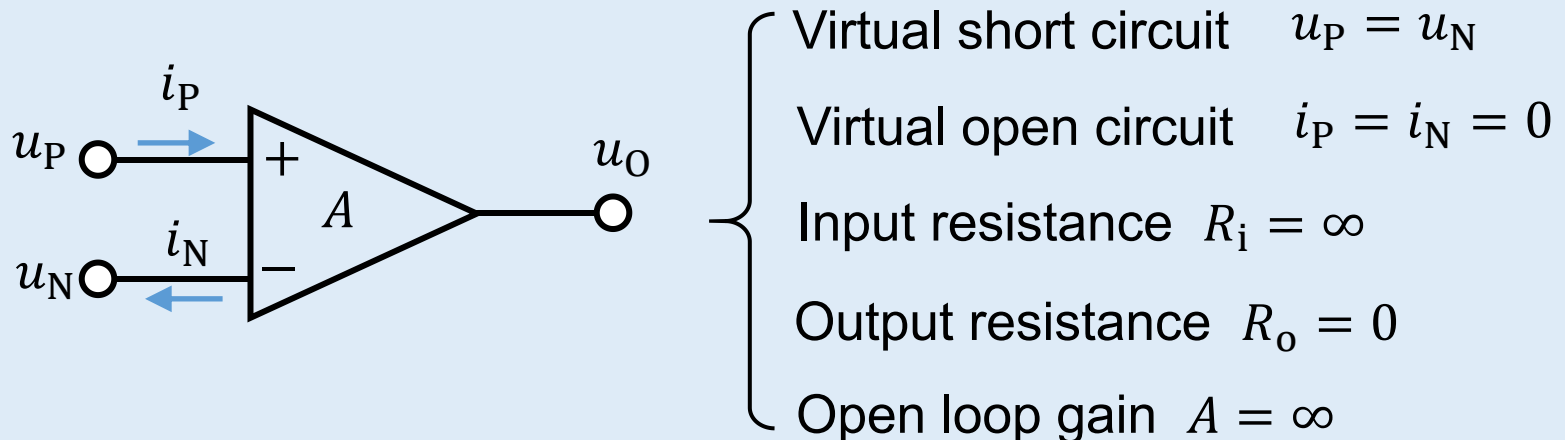
7.1 Basic signal operational circuits

Signal operation: The output voltage is a result of the operation of input voltage, such as addition, subtraction, division, multiplication, integration...

$$u_o = f(u_I)$$

Use **negative feedback** and **operational amplifier**

Ideal operational amplifier



7.2 Scaling operational circuit 比例运算电路

(1) Inverted input 反相输入

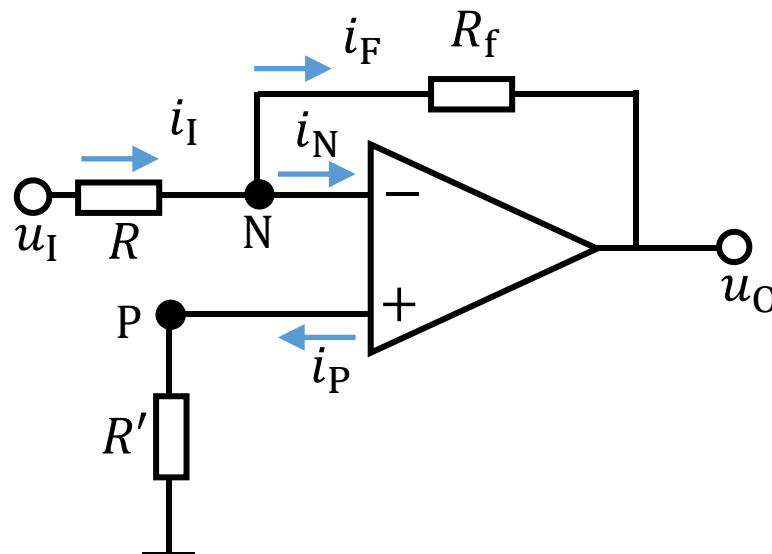
a. Which type feedback connection?

b. Voltage gain?

$$\dot{A}_u = \frac{u_O}{u_I} = \frac{-i_F R_f}{i_I R} = -\frac{R_f}{R}$$

c. Input resistance?

$$R_i = \frac{u_I}{i_I} = R$$



Virtual short circuit $u_P = u_N$

Virtual open circuit $i_P = i_N = 0$

(1) Inverted input 反相输入

d. The value of R' ?

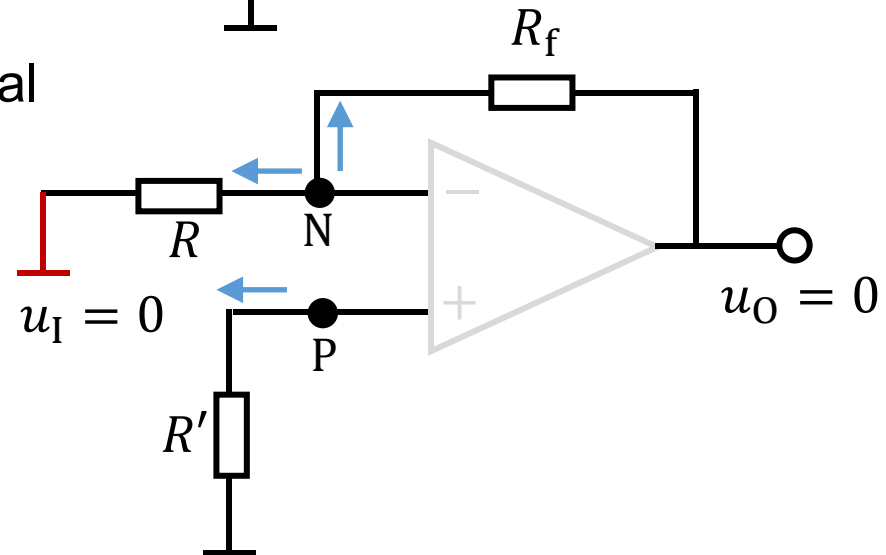
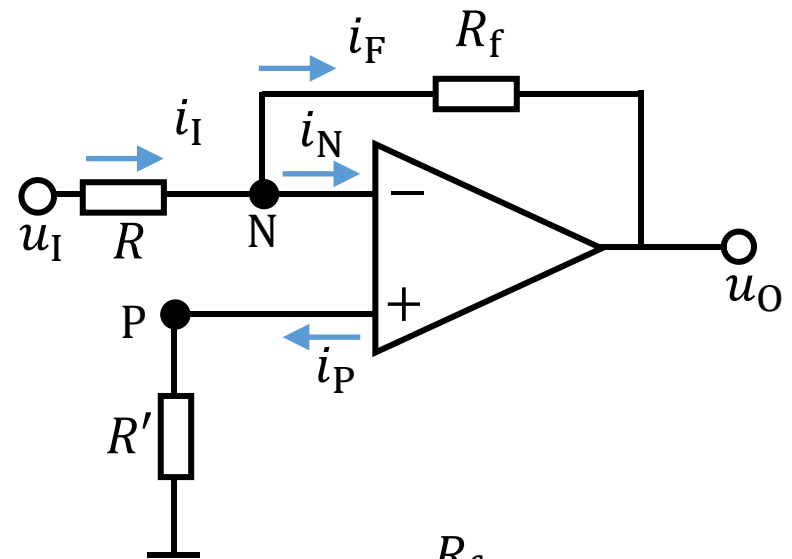
Keep the symmetry of the input circuit



The input resistance at P should equal to that at N



$$R' = R // R_f$$



(1) Inverted input 反相输入

e. The drawback of the circuit

When $\dot{A}_u = -100$, $R_i = 100 \text{ k}\Omega$:

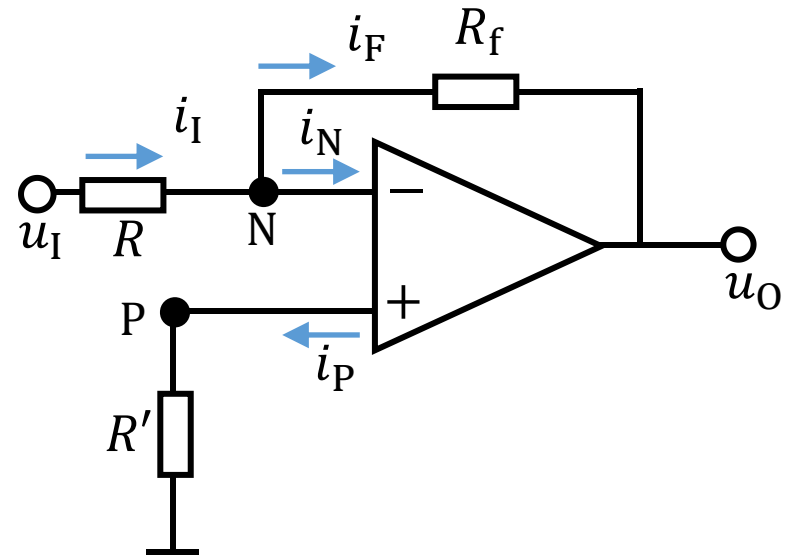


$$R_f = 10 \text{ M}\Omega$$



The noise and stability of resistor will be poor.

R_f will be at same order of the input resistance of operational amplifier, and thus $\dot{A}_u$ will change.



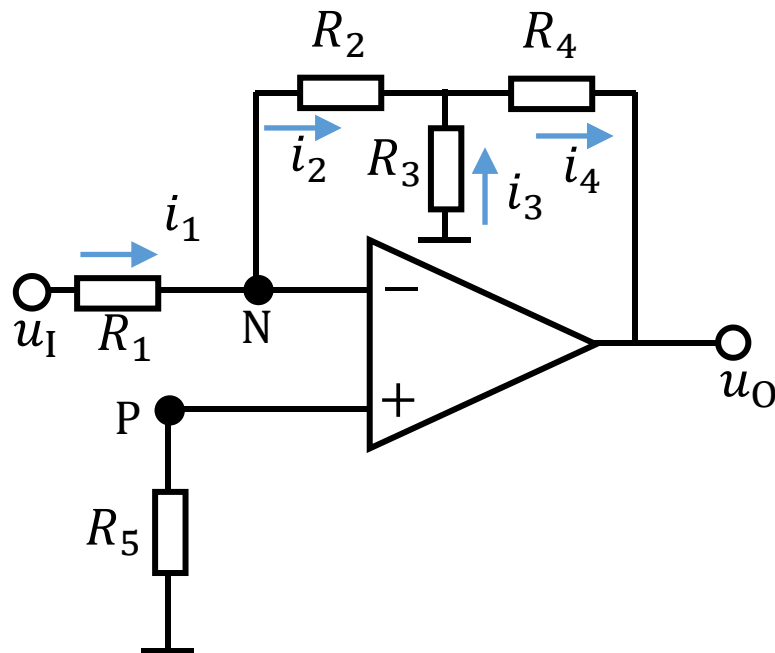
(2) T-shape feedback network inverted input T型反馈网络反相输入

a. Transmission characteristics 传输特性 ($u_O \sim u_I$):

$$\left\{ \begin{array}{l} u_O = -i_4(R_4 + R_2 // R_3) \\ u_I = i_1 R_1 \\ i_4 = i_2 \frac{R_2 + R_3}{R_3} = i_1 \frac{R_2 + R_3}{R_3} \end{array} \right.$$



$$u_O = -\frac{R_2 + R_4}{R_1} \left(1 + \frac{R_2 // R_4}{R_3}\right) u_I$$



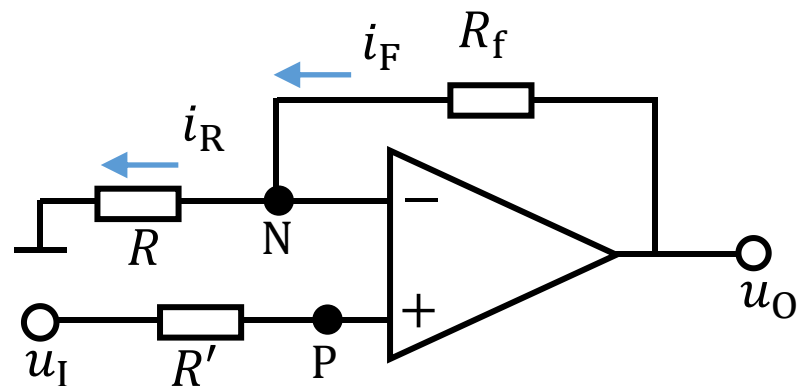
(3) Non-inverted input 同相输入

Voltage gain:

$$\left\{ \begin{array}{l} u_O = i_F R_f + i_R R \\ u_I = i_R R \\ i_F = i_R \end{array} \right.$$



$$u_O = \frac{R + R_f}{R} u_I$$

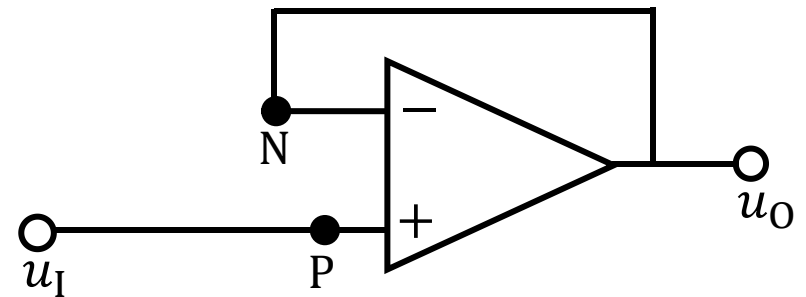
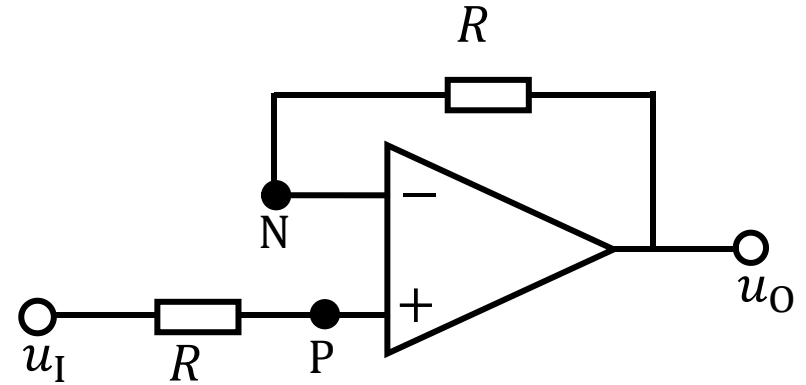


(3) Non-inverted input: Voltage follower 电压跟随器

$$\left\{ \begin{array}{l} u_O = u_N \\ u_I = u_P \\ u_P = u_N \end{array} \right.$$



$$u_O = u_I$$



7.3 Addition operation 加法运算

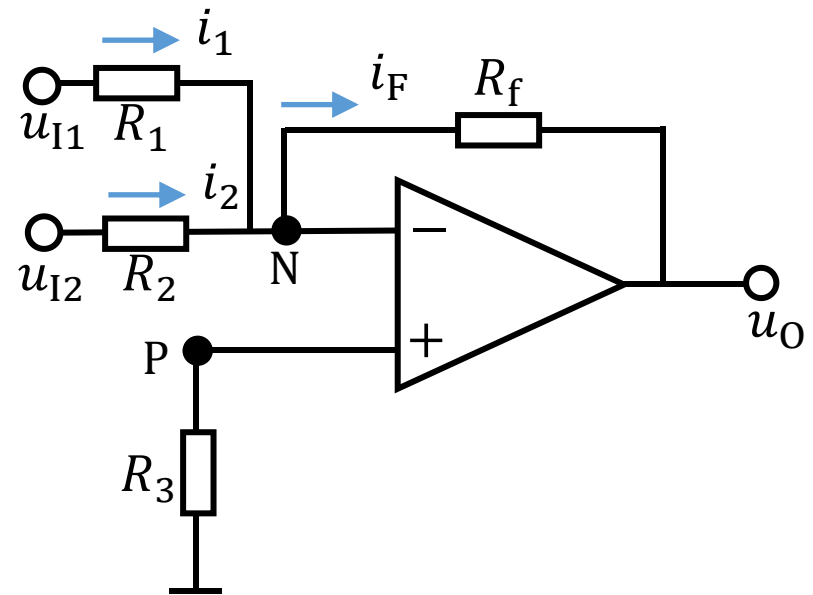
(1) Inverted input 反相输入

Output voltage $u_O = f(u_{I1}, u_{I2})$:

$$\left\{ \begin{array}{l} i_F = i_1 + i_2 \\ i_1 = \frac{u_{I1}}{R_1} \\ i_2 = \frac{u_{I2}}{R_2} \end{array} \right.$$

$$u_O = -i_F R_f$$

$$= -\left(\frac{R_f}{R_1} u_{I1} + \frac{R_f}{R_2} u_{I2}\right)$$



Virtual short circuit $u_P = u_N$

Virtual open circuit $i_P = i_N = 0$

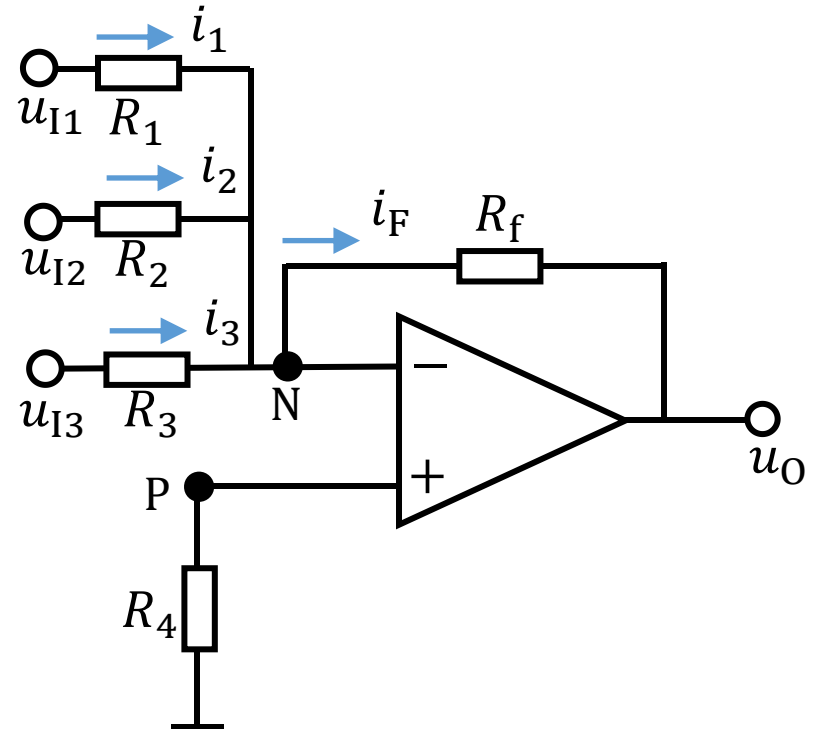
Output voltage $u_O = f(u_{I1}, u_{I2})$:

$$i_F = i_1 + i_2 + i_3$$

$$= \frac{u_{I1}}{R_1} + \frac{u_{I2}}{R_2} + \frac{u_{I3}}{R_3}$$

$$u_O = -i_F R_f$$

$$= -\left(\frac{R_f}{R_1} u_{I1} + \frac{R_f}{R_2} u_{I2} + \frac{R_f}{R_3} u_{I3}\right)$$



Superposition method 叠加法

a). Consider the contribution of u_{I1} :



Ground u_{I2} and u_{I3}

$$i_{F1} = i_1 = \frac{u_{I1}}{R_1} \quad u_{O1} = -\frac{R_f}{R_1} u_{I1}$$

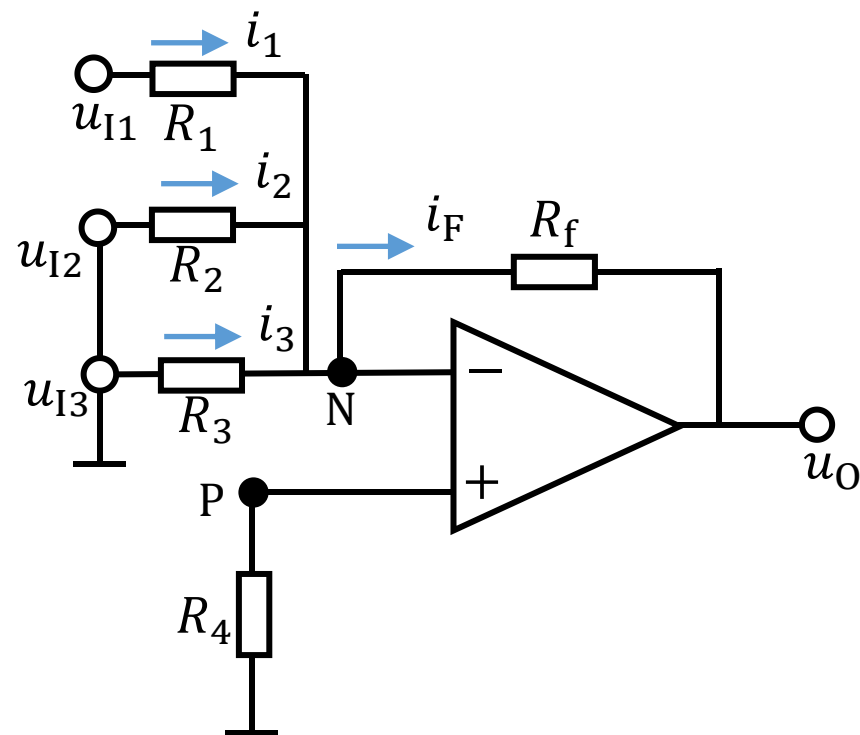
b). Consider the contribution of u_{I2} :



Ground u_{I1} and u_{I3}

$$i_{F2} = i_2 = \frac{u_{I2}}{R_2} \quad u_{O2} = -\frac{R_f}{R_2} u_{I2}$$

$$\begin{aligned} u_O &= u_{O1} + u_{O2} + u_{O3} \\ &= -\left(\frac{R_f}{R_1} u_{I1} + \frac{R_f}{R_2} u_{I2} + \frac{R_f}{R_3} u_{I3}\right) \end{aligned}$$



c). Consider the contribution of u_{I3} :



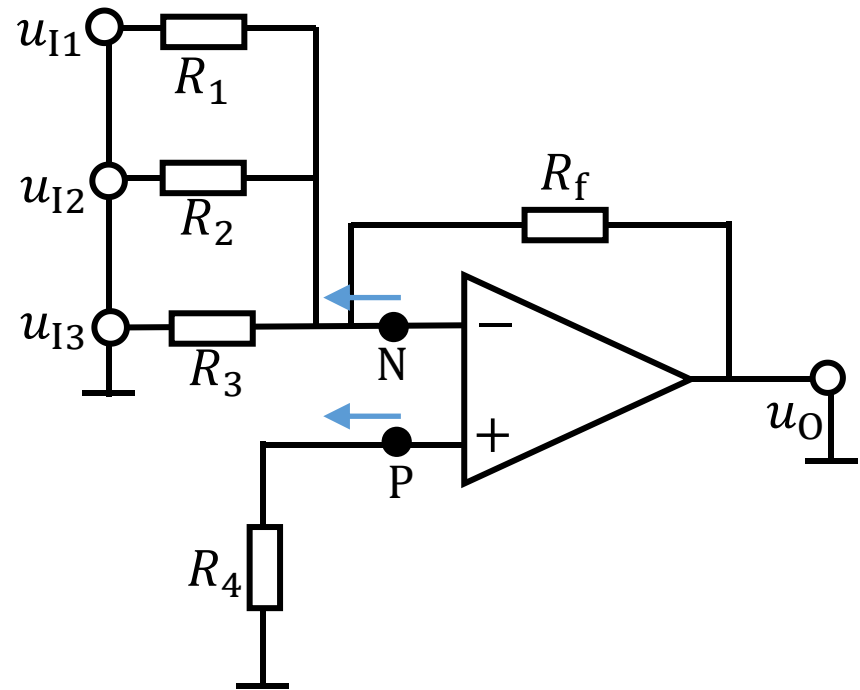
Ground u_{I1} and u_{I2}

$$i_{F3} = i_3 = \frac{u_{I3}}{R_3} \quad u_{O3} = -\frac{R_f}{R_3} u_{I3}$$

The value of R_4 ?

The symmetry of input circuit

$$R_4 = R_1 // R_2 // R_3 // R_f$$



(2) Non-inverted input 同相输入

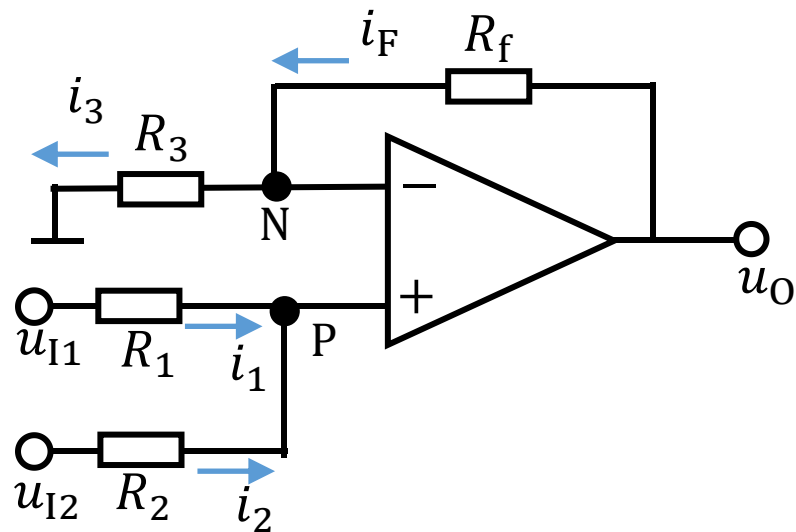
Output voltage $u_O = f(u_{I1}, u_{I2})$:

$$\begin{cases} i_F = i_3 \\ i_1 = -i_2 \end{cases}$$

$$\frac{u_{I1} - u_P}{R_1} = -\frac{u_{I2} - u_P}{R_2}$$

$$\begin{aligned} & u_P = u_N \\ & = \frac{R_2}{R_1 + R_2} u_{I1} + \frac{R_1}{R_1 + R_2} u_{I2} \end{aligned}$$

$$= \frac{R_2}{R_1 + R_2} \frac{u_{I1}}{R_3} + \frac{R_1}{R_1 + R_2} \frac{u_{I2}}{R_3}$$



$$\begin{aligned} u_O &= i_F R_f + i_3 R_3 \\ &= \frac{R_3 + R_f}{R_3} (K_1 u_{I1} + K_2 u_{I2}) \end{aligned}$$

$$K_1 = \frac{R_2}{R_1 + R_2}, K_2 = \frac{R_1}{R_1 + R_2}$$

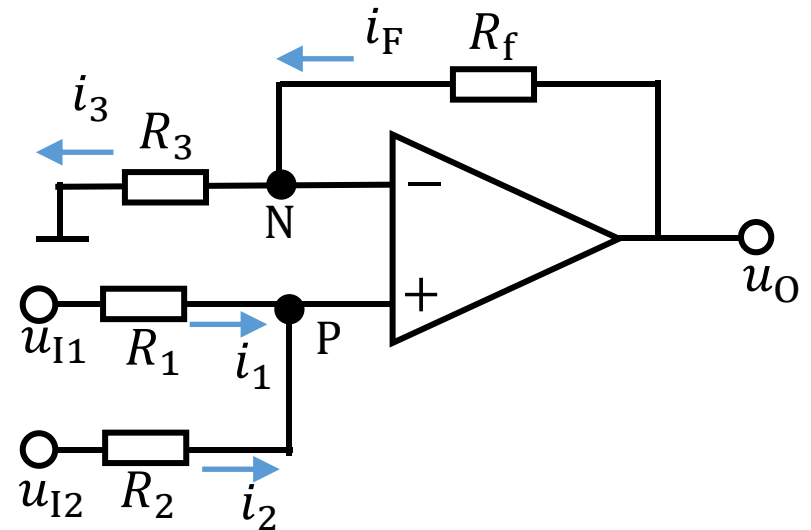
The symmetry of input circuit



$$R_3 // R_f = R_1 // R_2$$



$$u_O = \frac{R_f}{R_1} u_{I1} + \frac{R_f}{R_2} u_{I2}$$



7.4 Subtraction operation 减法运算

Output voltage $u_O = f(u_{I1}, u_{I2})$:

$$i_2 = i_3$$

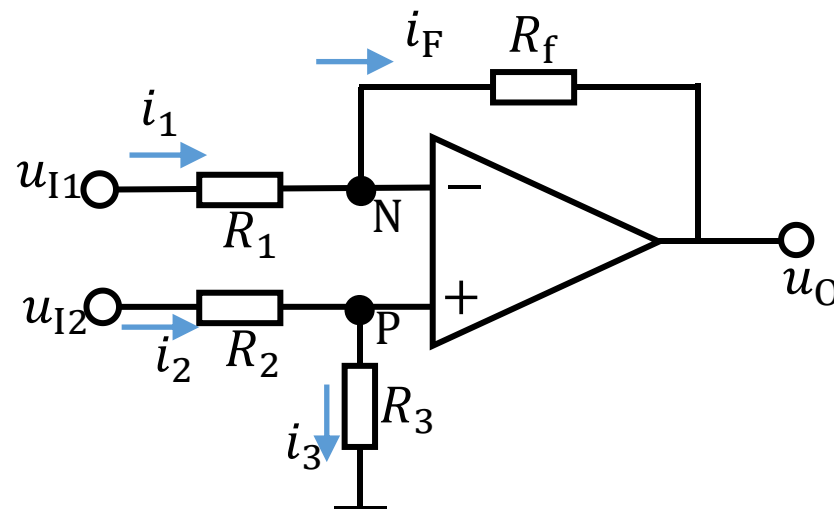
$$u_N = u_P = \frac{R_3}{R_2 + R_3} u_{I2}$$

$$i_F = i_1$$

$$= \frac{u_{I1} - u_N}{R_1}$$

$$u_O = u_N - i_F R_f$$

$$= -\frac{R_f}{R_1} u_{I1} + \frac{R_1 + R_f}{R_1} \frac{R_3}{R_2 + R_3} u_{I2}$$



The symmetry of input circuit



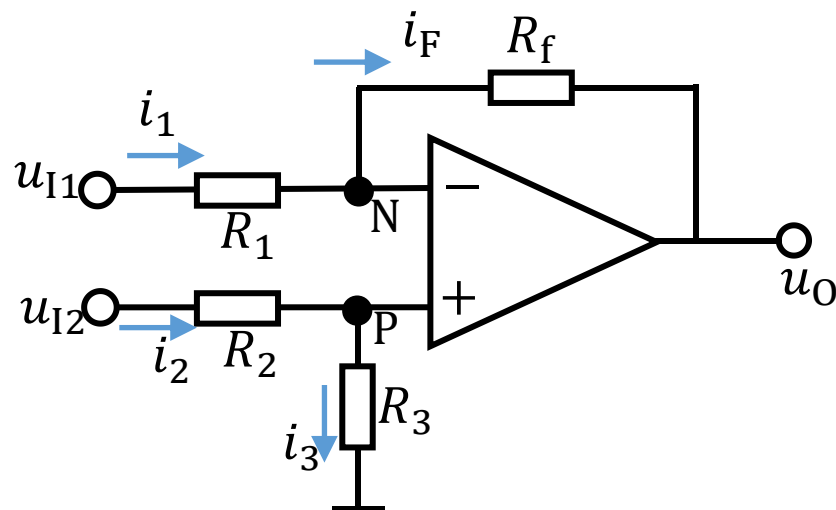
$$R_2 // R_3 = R_1 // R_f$$



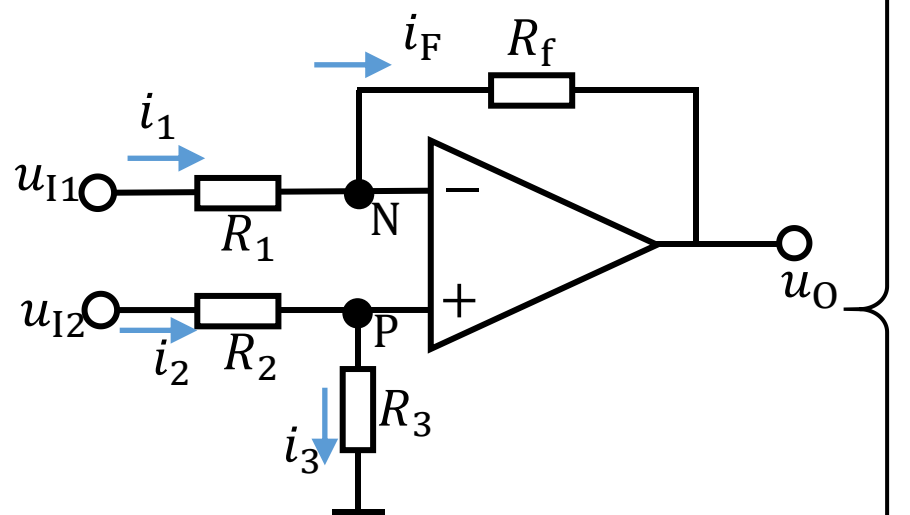
$$u_O = -\frac{R_f}{R_1} u_{I1} + \frac{R_f}{R_2} u_{I2}$$

Special case: $\begin{cases} R_2 = R_1 \\ R_3 = R_f \end{cases}$

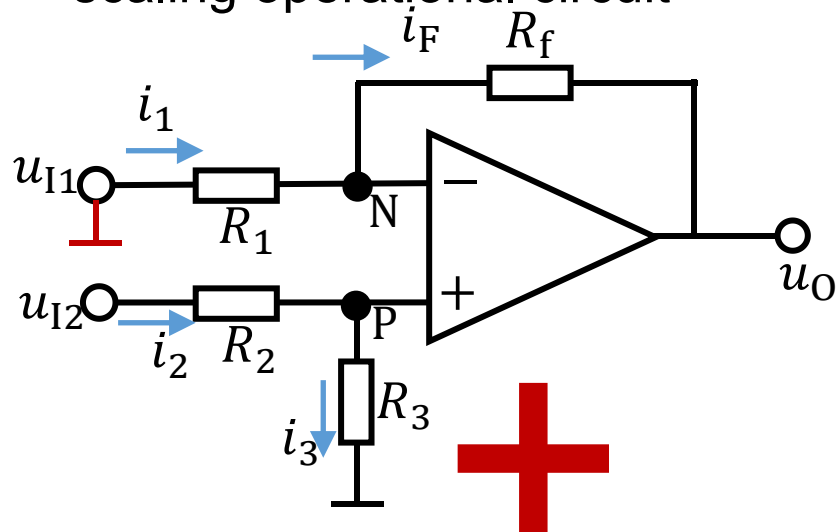
$$u_O = \frac{R_f}{R_1} (u_{I2} - u_{I1})$$



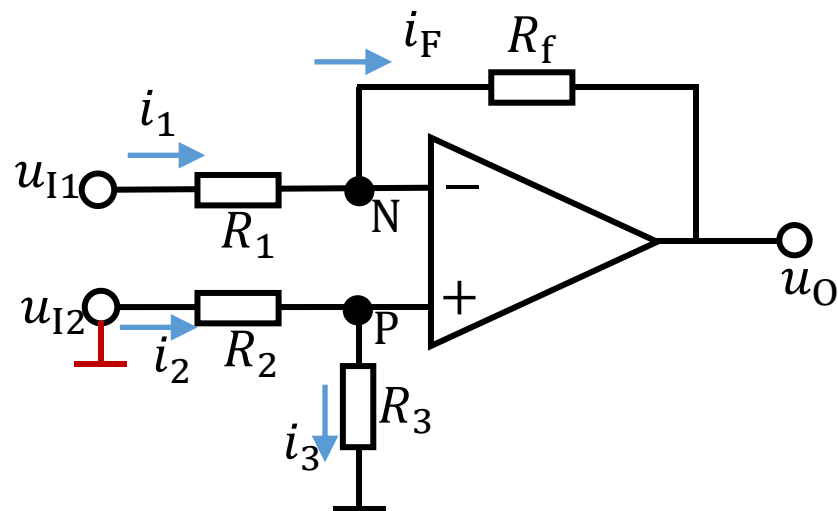
Superposition method 叠加法



Non-inverted input
scaling operational circuit



Inverted input scaling
operational circuit

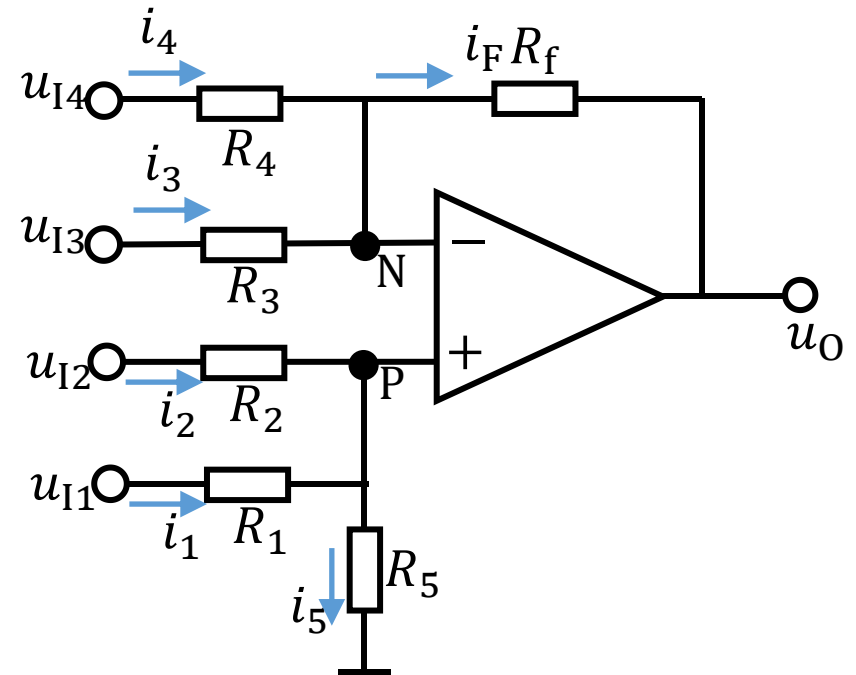


7.5 Addition and subtraction mixed operation 加减法混合运算

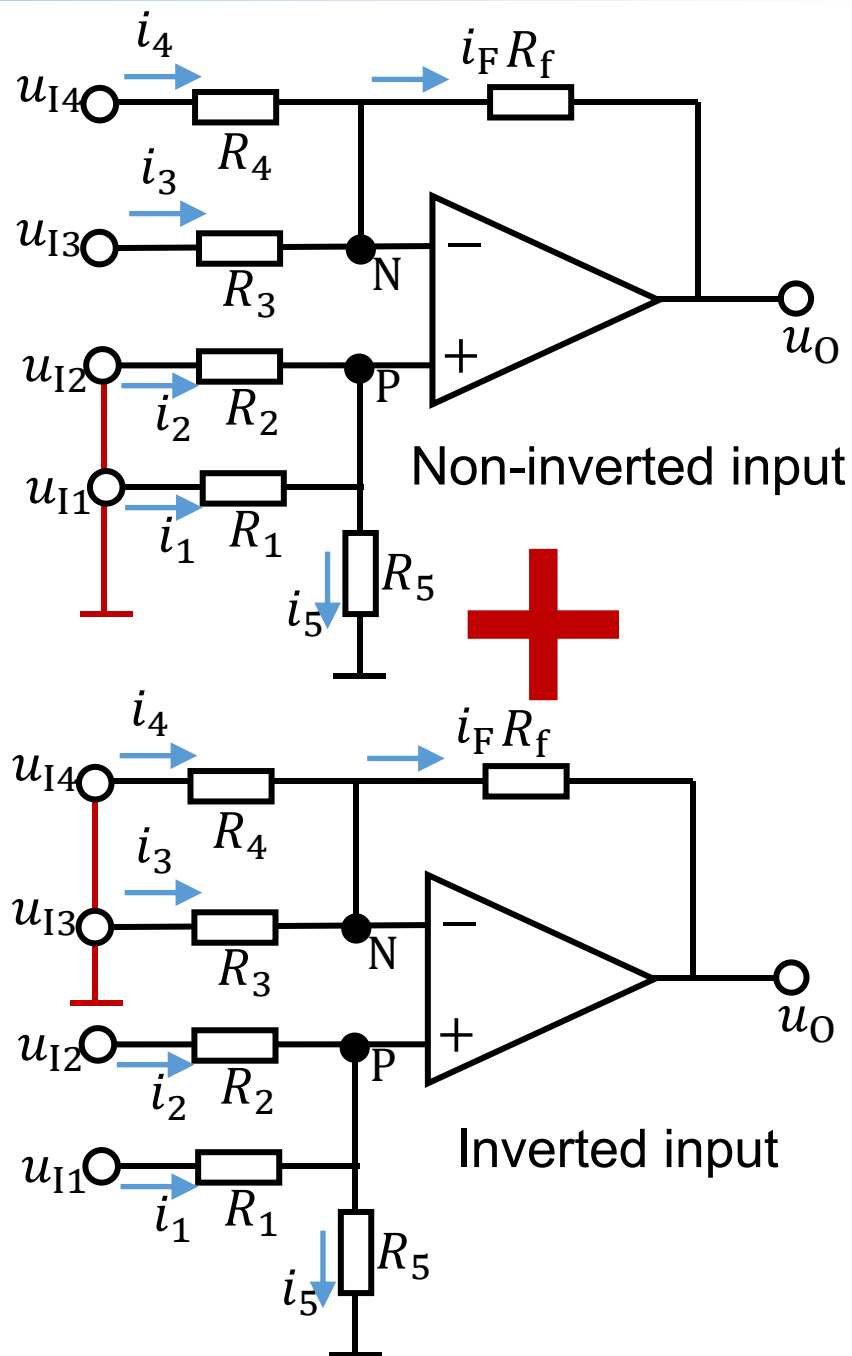
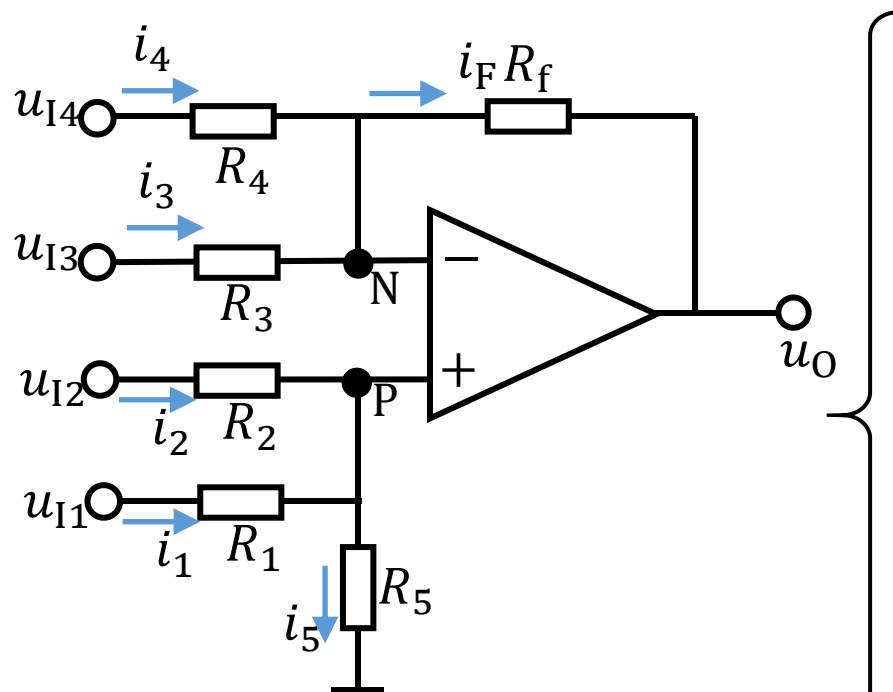
How to realize

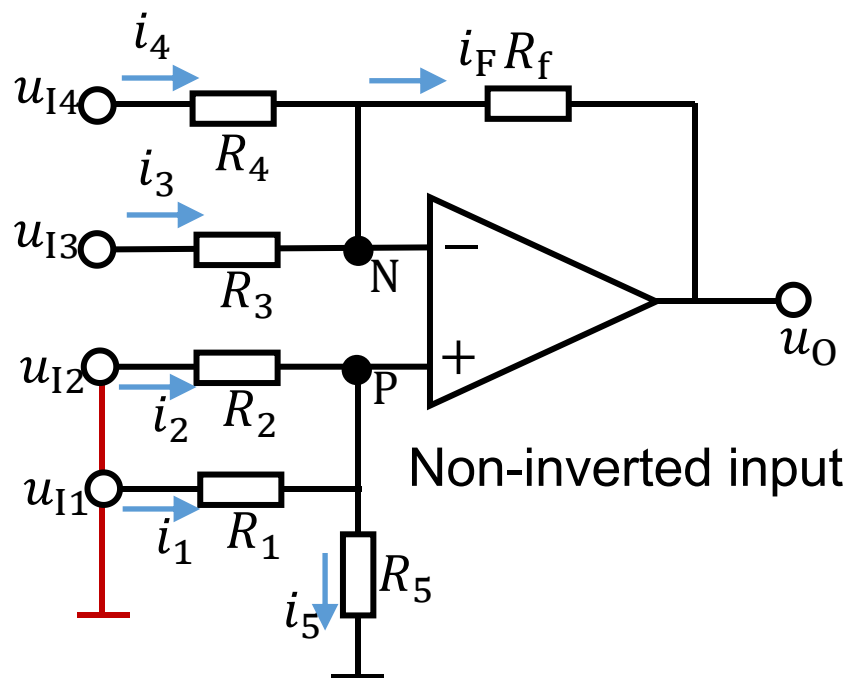
$$u_O = 10u_{I1} + 5u_{I2} - 20u_{I3} - 4u_{I4}?$$

Non-inverted input **Inverted input**

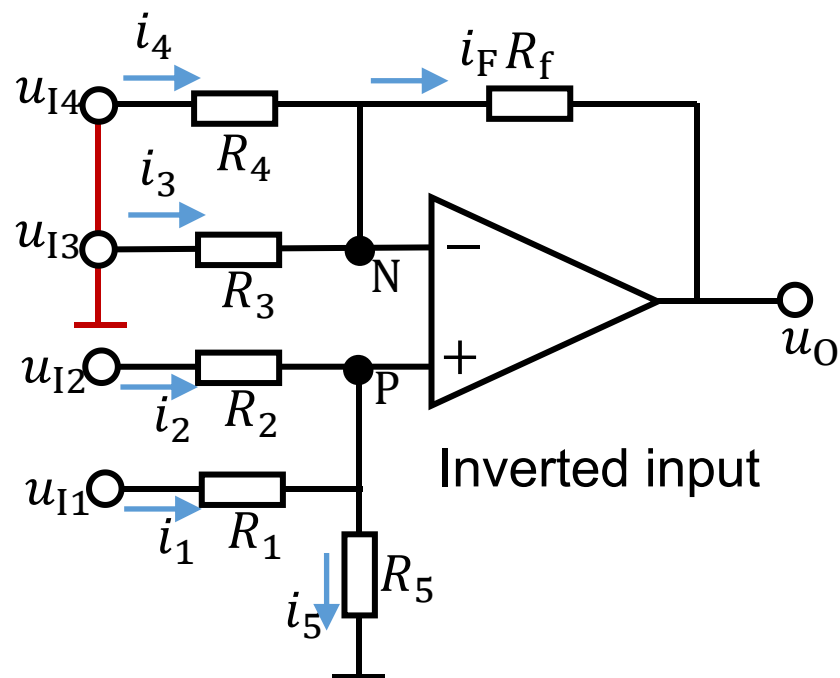


Superposition method 叠加法





$$u_{O1} = - \left(\frac{R_f}{R_3} u_{I3} + \frac{R_f}{R_4} u_{I4} \right)$$



$$\begin{cases} \frac{u_{I1} - u_P}{R_1} + \frac{u_{I2} - u_P}{R_2} = \frac{u_P}{R_5} \\ R_1 // R_2 // R_5 = R_3 // R_4 // R_f \end{cases}$$

$$u_{O2} = \left(\frac{R_f}{R_1} u_{I1} + \frac{R_f}{R_2} u_{I2} \right)$$

$$u_O = u_{O1} + u_{O2}$$

$$= \frac{R_f}{R_1} u_{I1} + \frac{R_f}{R_2} u_{I2} - \frac{R_f}{R_3} u_{I3} - \frac{R_f}{R_4} u_{I4}$$

$$R_1 // R_2 // R_5 = R_3 // R_4 // R_f$$

How to realize

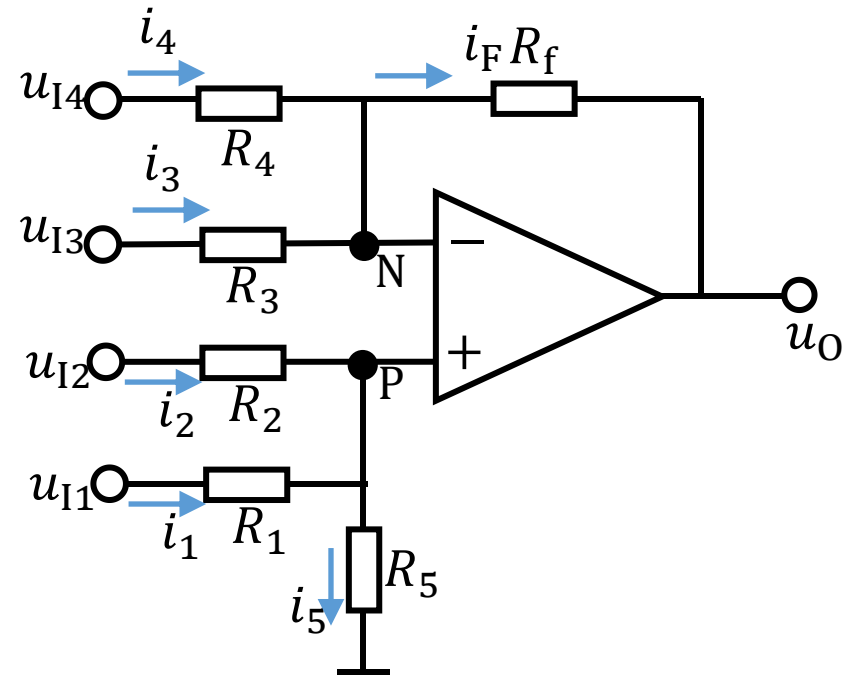
$$u_O = 10u_{I1} + 5u_{I2} - 20u_{I3} - 4u_{I4}?$$

When $R_f = 100\text{k}\Omega$:

$$R_1 = 10\text{k}\Omega, R_2 = 20\text{k}\Omega,$$

$$R_3 = 5\text{k}\Omega, R_4 = 25\text{k}\Omega$$

$$R_5 = \frac{500R_f}{500 + 45R_f} = 10\text{k}\Omega$$



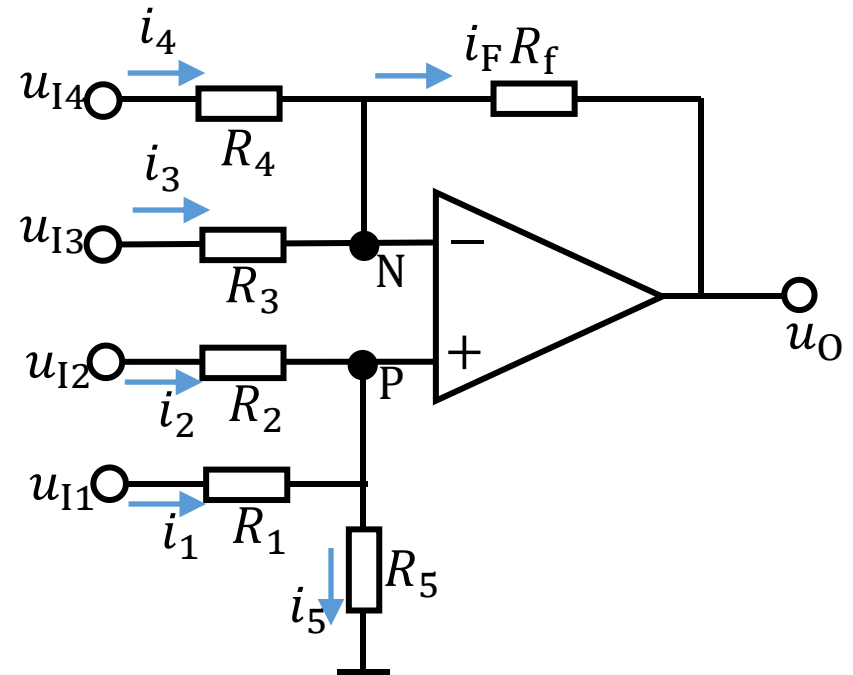
$$u_O = \frac{R_f}{R_1} u_{I1} + \frac{R_f}{R_2} u_{I2} - \frac{R_f}{R_3} u_{I3} - \frac{R_f}{R_4} u_{I4}$$

$$R_1 // R_2 // R_5 = R_3 // R_4 // R_f$$

Can this circuit realize

$$u_O = Au_{I1} + Bu_{I2} - Cu_{I3} - Du_{I4}$$

with A , B , C and $D > 0$?



$$u_O = \frac{R_f}{R_1} u_{I1} + \frac{R_f}{R_2} u_{I2} - \frac{R_f}{R_3} u_{I3} - \frac{R_f}{R_4} u_{I4}$$

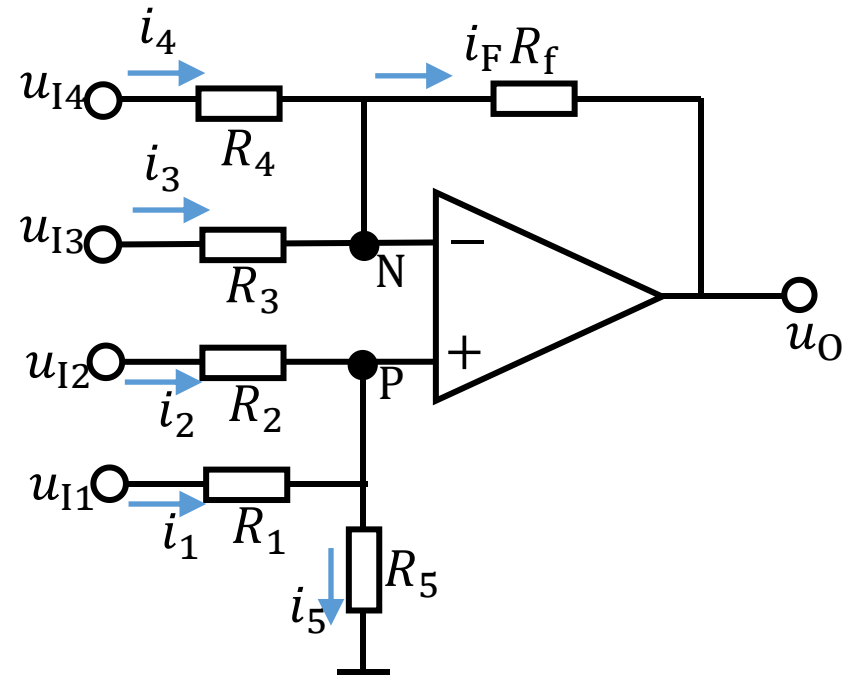
$$R_1 // R_2 // R_5 = R_3 // R_4 // R_f$$

$$u_O = 10u_{I1} + 5u_{I2} - 2u_{I3} - 4u_{I4}?$$

When $R_f = 100\text{k}\Omega$:

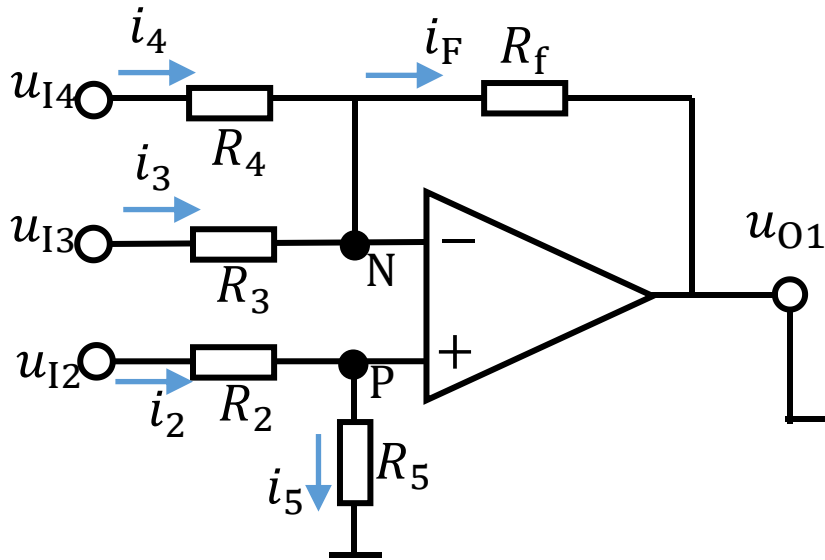
$$R_1 = 10\text{k}\Omega, R_2 = 20\text{k}\Omega, \\ R_3 = 50\text{k}\Omega, R_4 = 25\text{k}\Omega$$

$$R_5 = \frac{100R_f}{100 - 9R_f} < 0$$



$u_O = 10u_{I1} + 5u_{I2} - 2u_{I3} - 4u_{I4}$ cannot be realized for this circuit!

$$u_O = 10u_{I1} + 5u_{I2} - 2u_{I3} - 4u_{I4}?$$

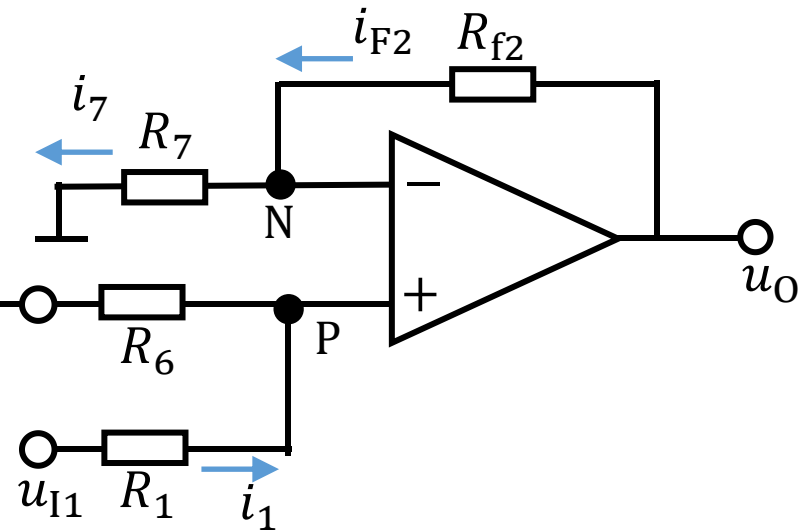


$$\begin{cases} u_{O1} = \frac{R_f}{R_2} u_{I2} - \frac{R_f}{R_3} u_{I3} - \frac{R_f}{R_4} u_{I4} \\ R_2 // R_5 = R_3 // R_4 // R_f \end{cases}$$

When $R_f = 100\text{k}\Omega$:

$$R_2 = 20\text{k}\Omega, R_3 = 50\text{k}\Omega, R_4 = 25\text{k}\Omega$$

$$R_5 = \frac{100R_f}{100 + R_f} = 50\text{k}\Omega$$



$$\begin{cases} R_7 // R_{f2} = R_1 // R_6 \\ u_O = \frac{R_{f2}}{R_6} u_{O1} + \frac{R_{f2}}{R_1} u_{I1} \end{cases}$$

When $R_{f2} = 100\text{k}\Omega$:

$$R_6 = 100\text{k}\Omega, R_1 = 10\text{k}\Omega, R_7 = 10\text{k}\Omega$$

7.6 Integration operation 积分运算

$$\left\{ \begin{array}{l} i_C = i_I = \frac{u_I}{R} \\ i_C = \frac{dQ}{dt} = C \frac{du_C}{dt} \end{array} \right.$$

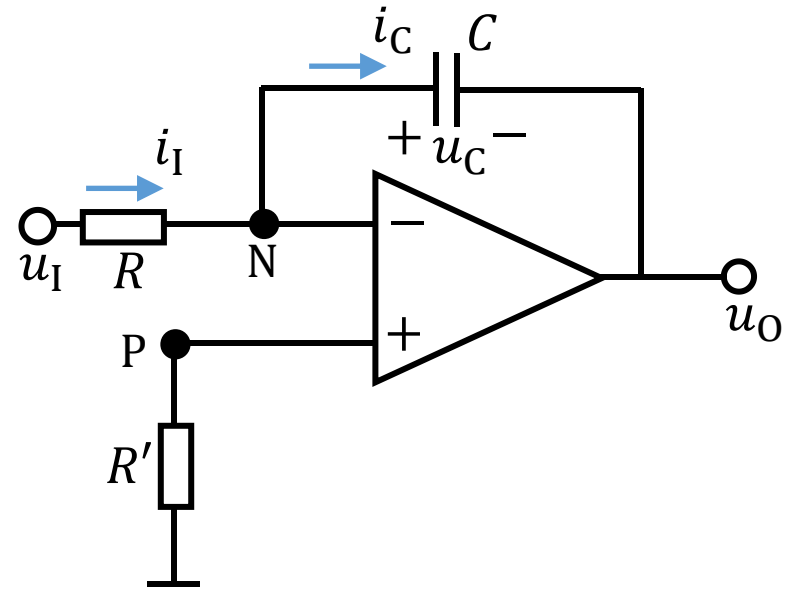


$$u_O = -u_C$$

=

$$-\frac{1}{C} \int i_C dt$$

$$= -\frac{1}{RC} \int u_I dt$$

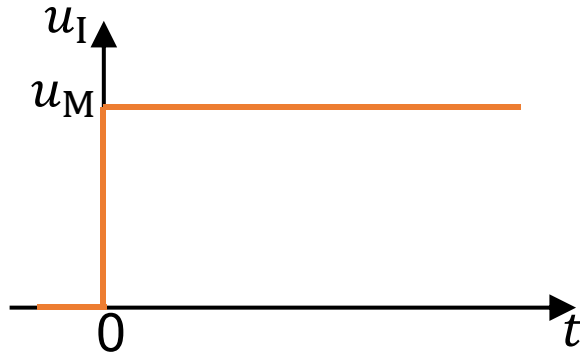


From time t_1 to t_2 :

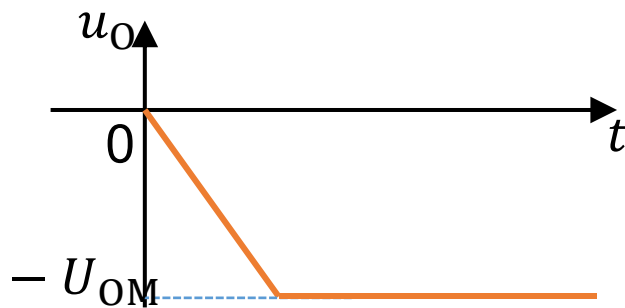
$$u_O = -\frac{1}{RC} \int_{t_1}^{t_2} u_I dt + u_O(t_1)$$

$$u_O = -\frac{1}{RC} \int_{t_1}^{t_2} u_I dt + u_O(t_1)$$

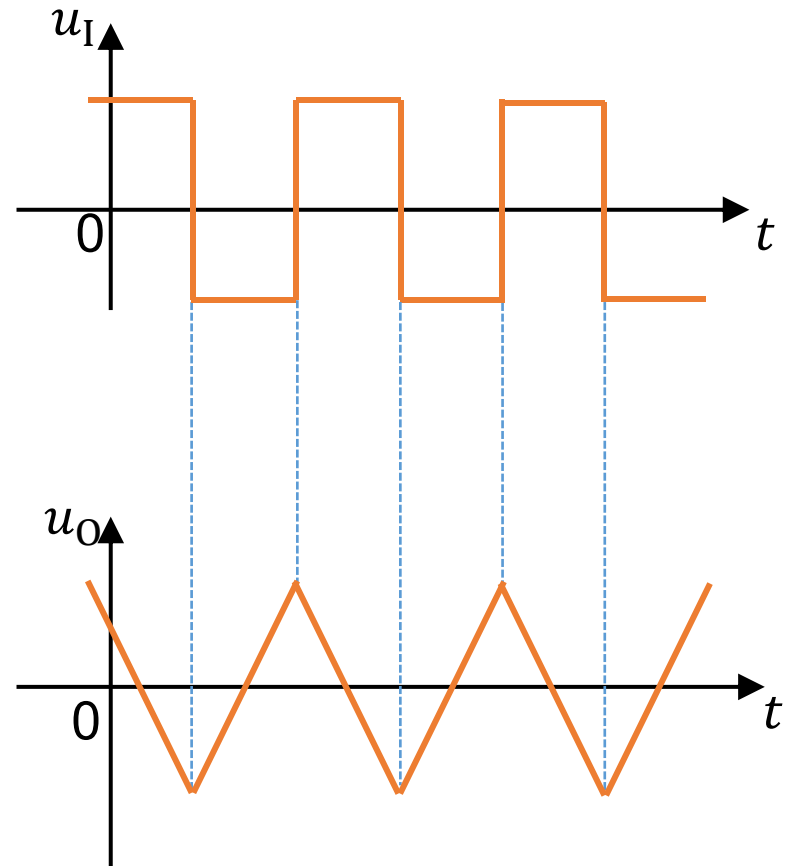
(1) Step signal 阶跃信号



$$u_O = -\frac{1}{RC} u_M t$$



(2) Square to triangular wave transition



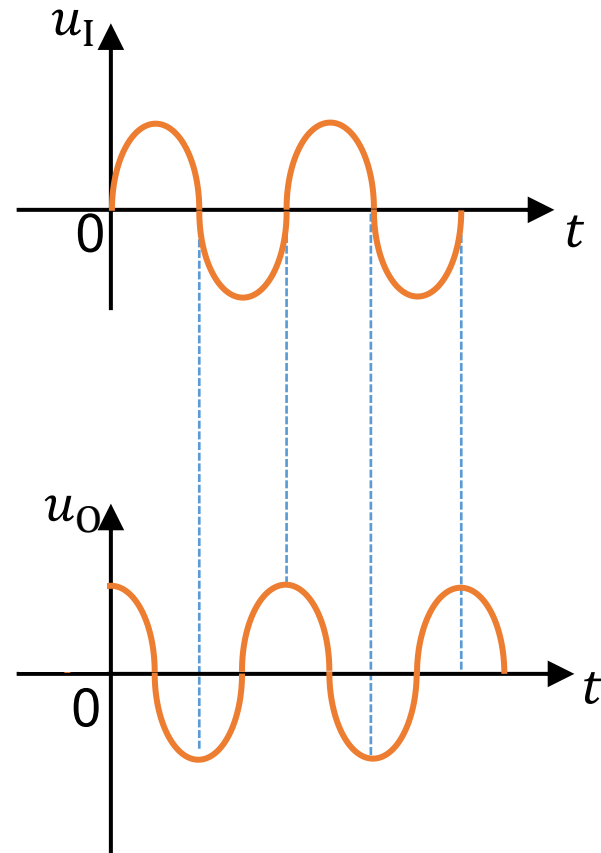
$$u_O = -\frac{1}{RC} \int_{t_1}^{t_2} u_I dt + u_O(t_1)$$

$$\left\{ \begin{array}{l} u_I = \sin(\omega t) \quad u_O(0) = \frac{1}{\omega RC} \\ \int \sin(\omega t) dt = -\frac{1}{\omega} \cos(\omega t) \end{array} \right.$$



$$u_O = \frac{1}{\omega RC} \cos(\omega t)$$

(3) Sin to Cos wave transition
(Phase shift 180°)



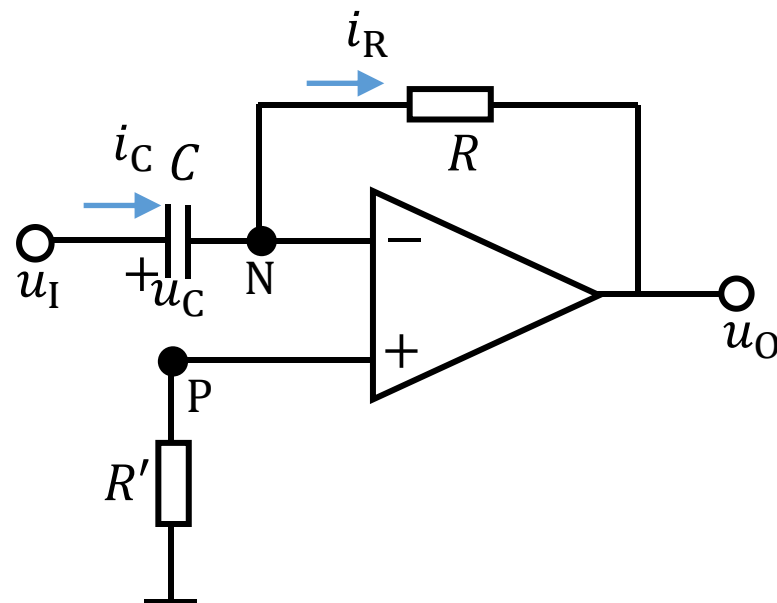
7.7 Differential operation 微分运算

$$\left\{ \begin{array}{l} i_R = i_C \\ i_C = \frac{dQ}{dt} = C \frac{du_C}{dt} = C \frac{du_I}{dt} \end{array} \right.$$

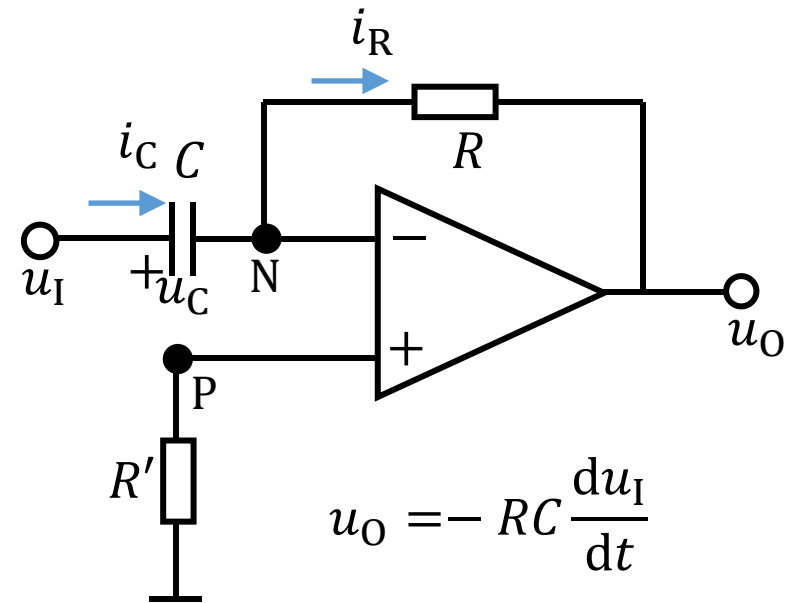


$$u_O = -i_R R$$

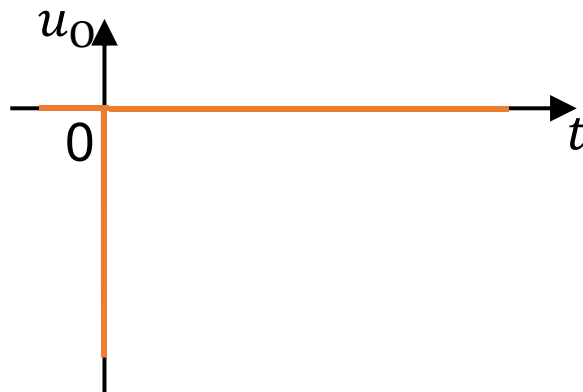
$$= -RC \frac{du_I}{dt}$$



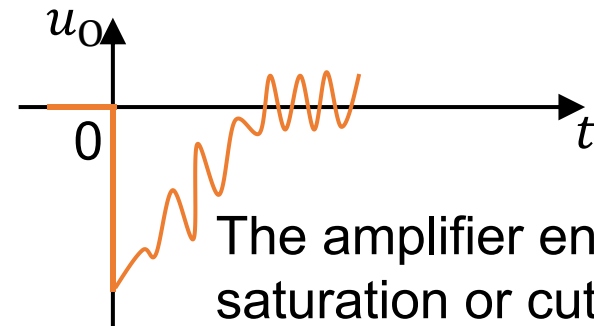
(1) Step signal



We expect:

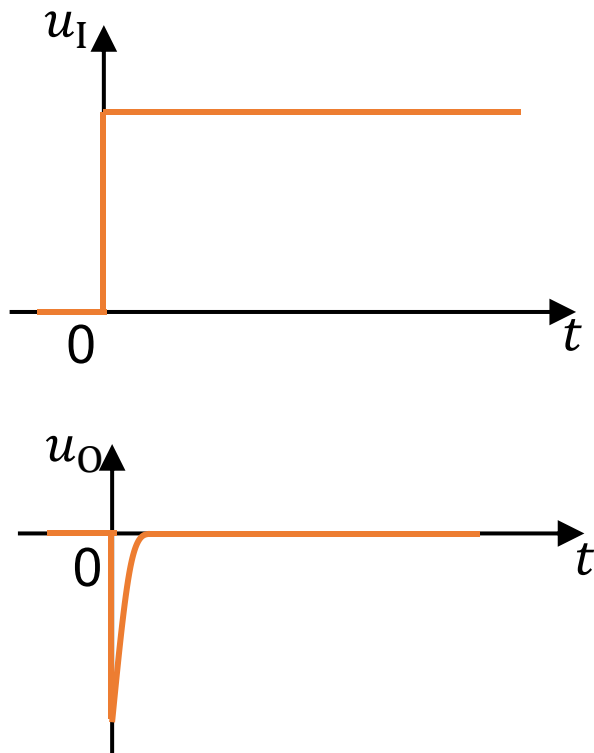


Real case:



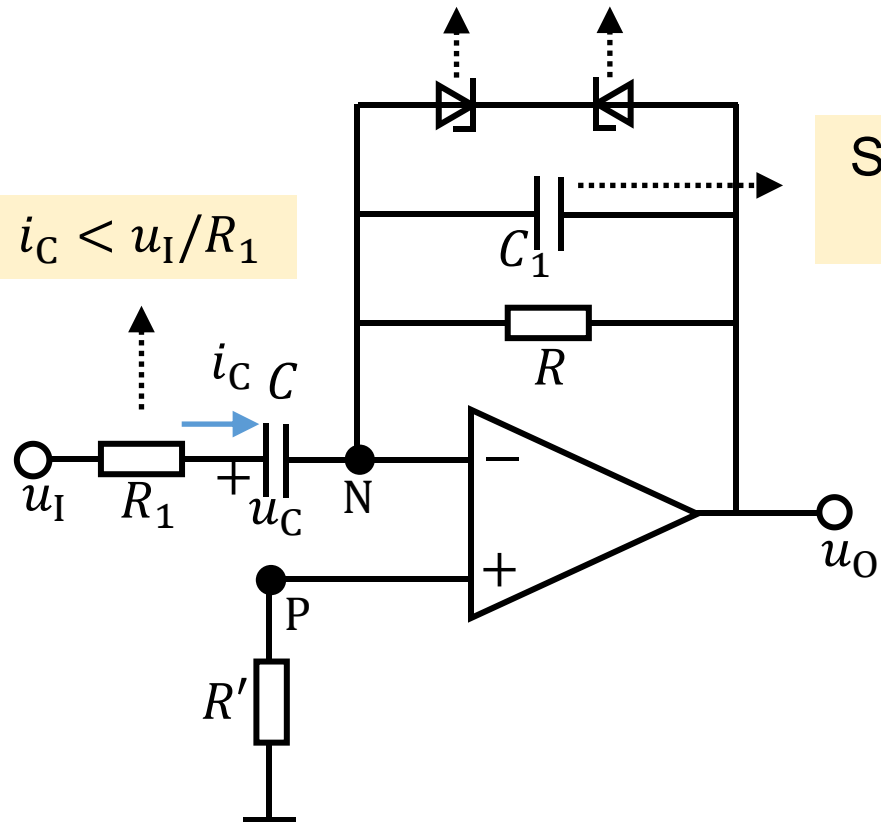
The amplifier enters saturation or cut-off region

Applicable differential operation



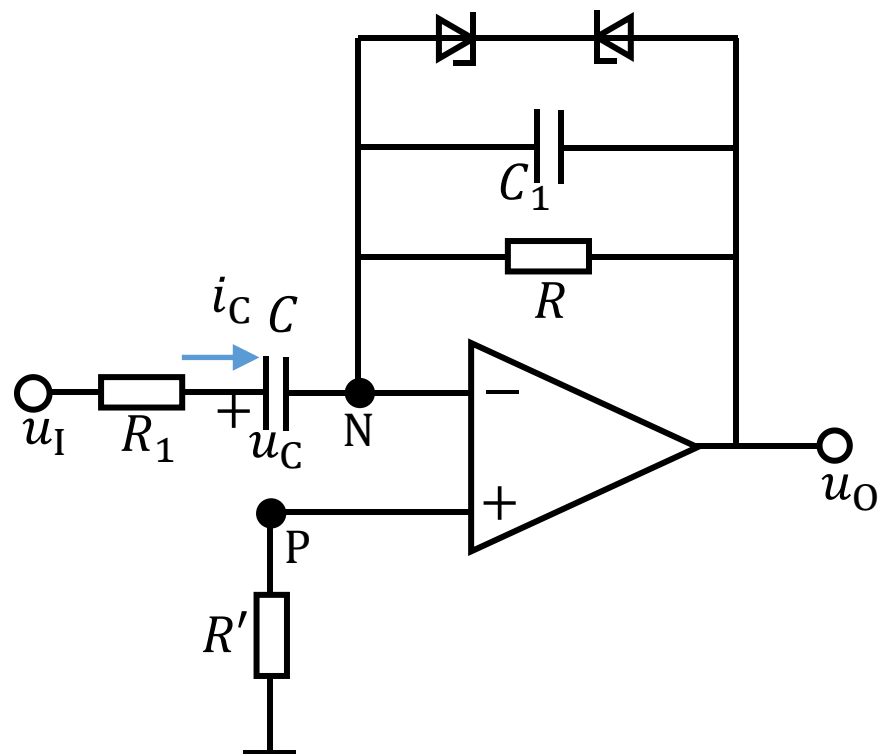
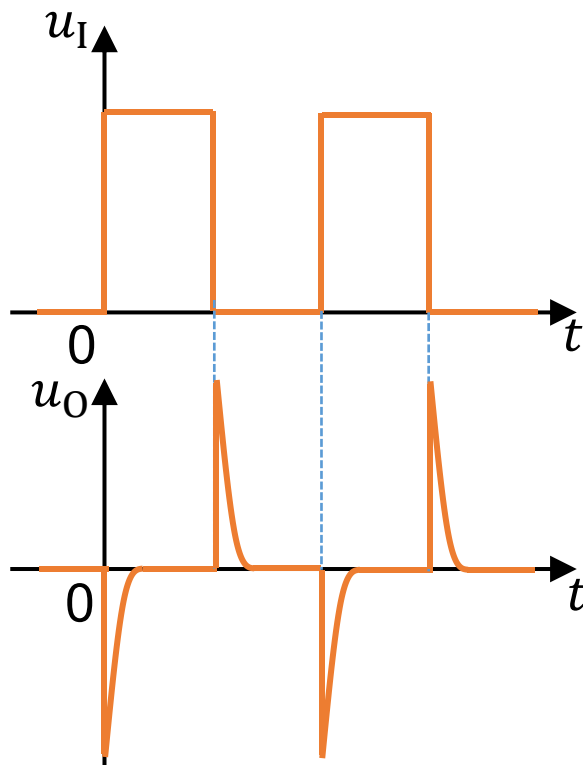
Restrict $u_{O-\max}$ to $\sim U_Z$

$$i_C < u_I / R_1$$



Stabilize
circuit

(2) Square wave



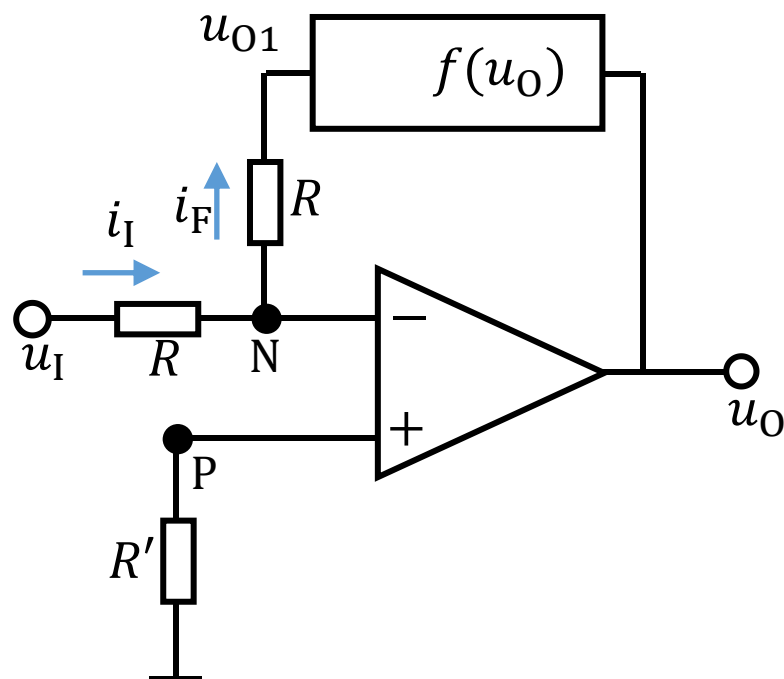
Use inverse function to realize differential operation

用逆函数实现微分运算

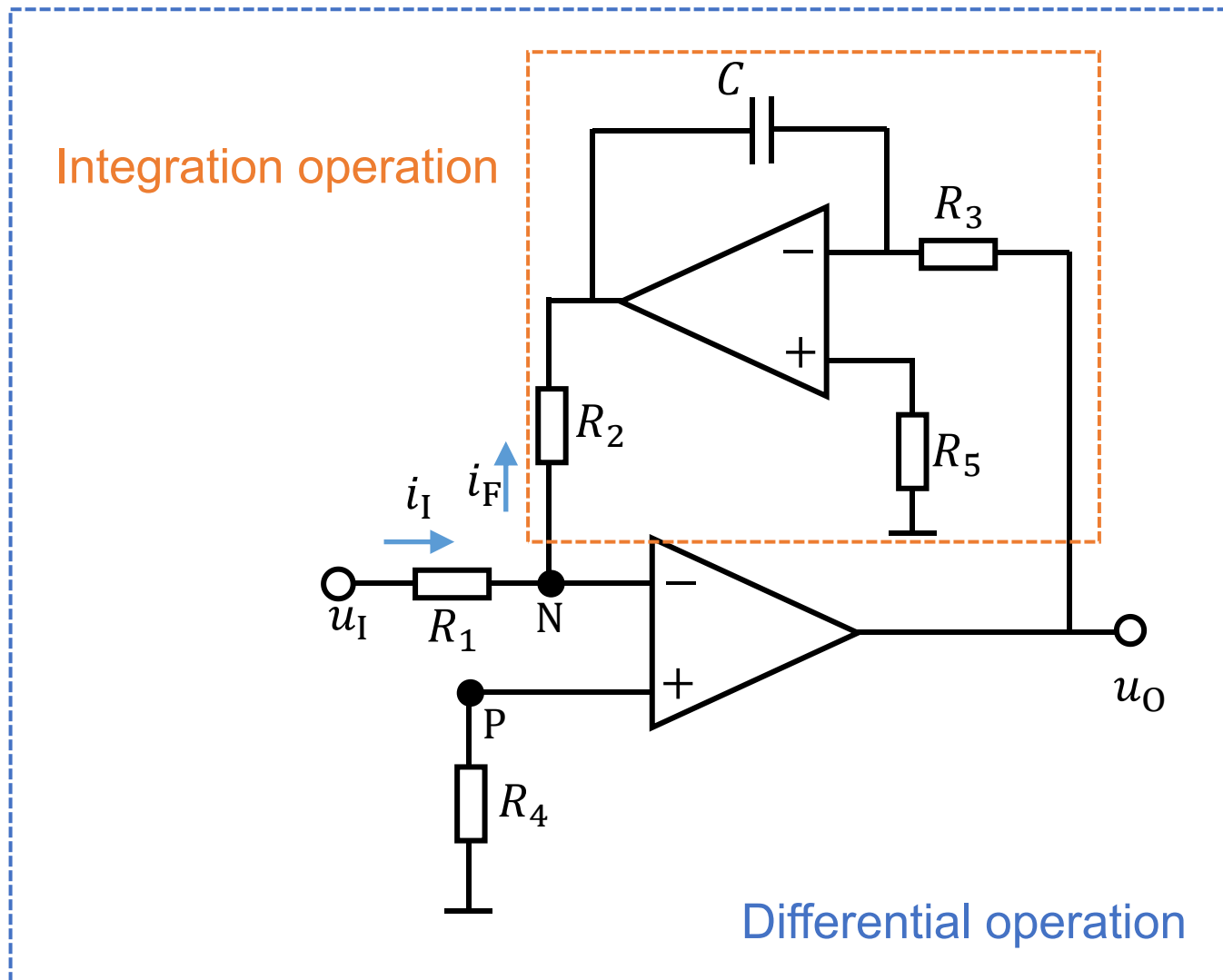
$$u_{O1} = f(u_O)$$

$$u_{O1} = -i_F R = -u_I$$

$$\left\{ \begin{array}{l} -u_I = f(u_O) \\ u_O = f^{-1}(-u_I) \end{array} \right.$$



Use inverse function to realize differential operation



7.8 Logarithm operation 对数运算

(1) Use diode

$$i_D \approx I_S e^{\frac{u_D}{U_T}}$$

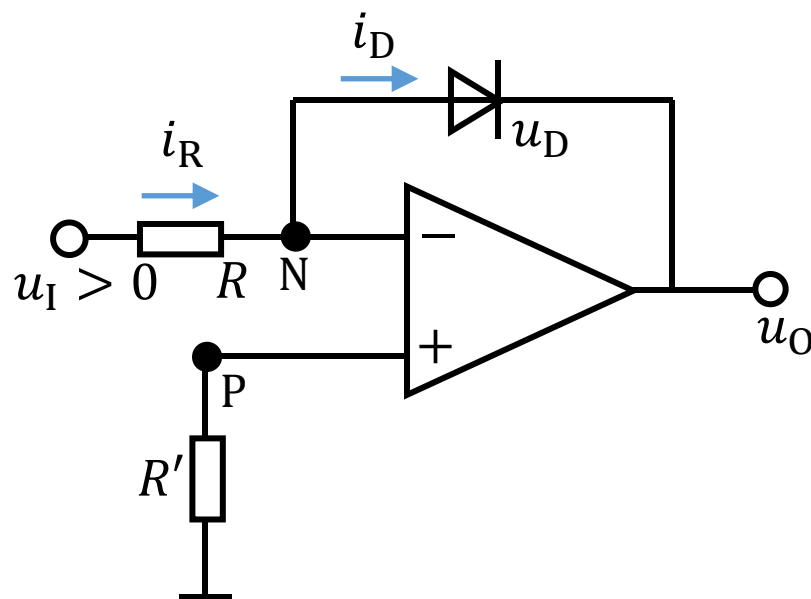


$$u_D \approx U_T \ln \frac{i_D}{I_S}$$

$$\because i_D = i_R = \frac{u_I}{R}$$



$$u_O = -u_D \approx -U_T \ln \frac{u_I}{I_S R}$$



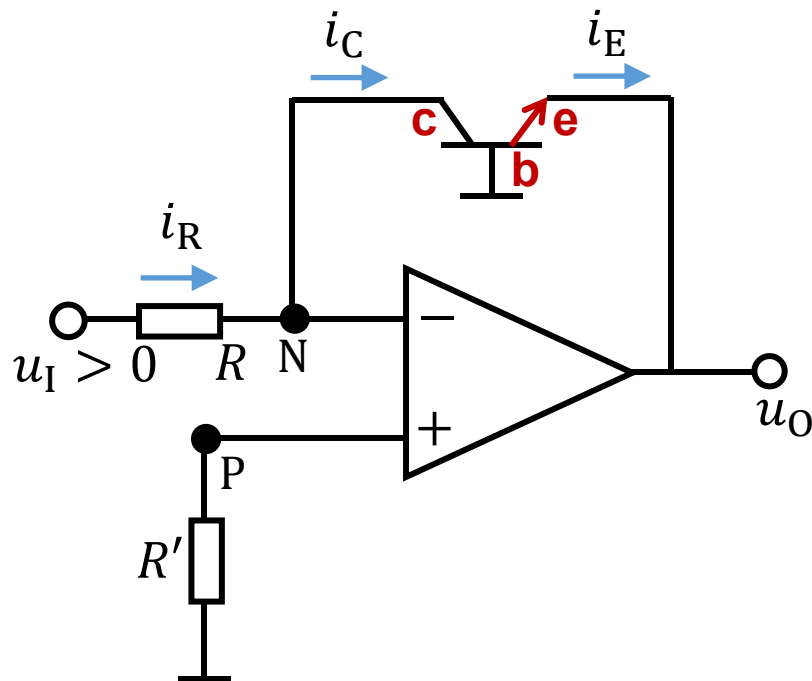
(2) Use transistor

$$\left\{ \begin{array}{l} i_C = i_R = \frac{u_I}{R} \\ i_C \approx i_E \approx I_S e^{\frac{u_{BE}}{U_T}}, u_{BE} \gg U_T \end{array} \right.$$



$$u_O = -u_{BE}$$

$$\approx -U_T \ln \frac{u_I}{I_S R}$$



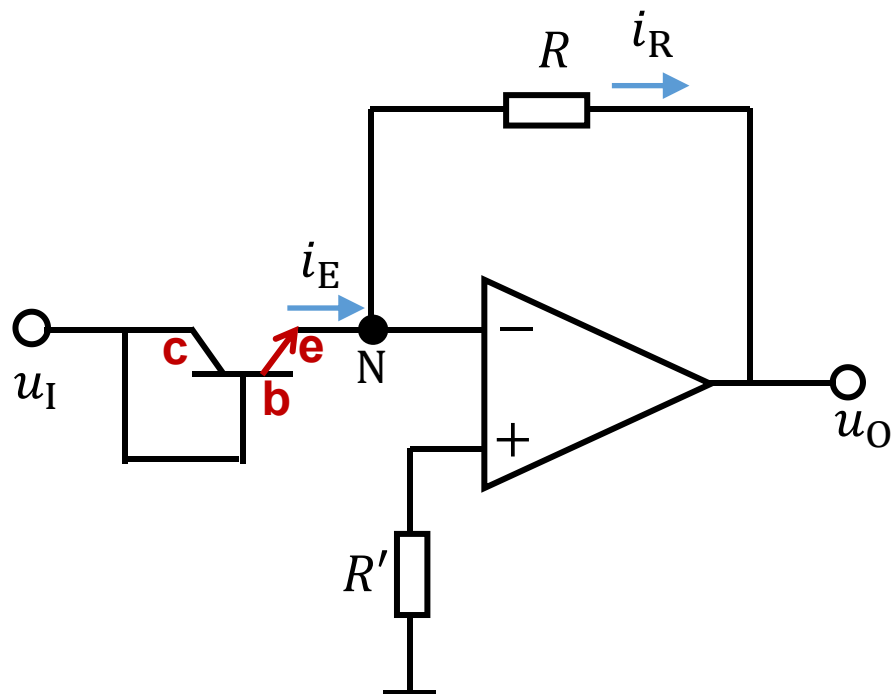
7.9 Exponent operation 指数运算

$$\left\{ \begin{array}{l} u_I = u_{BE} \\ i_R = i_E \approx I_S e^{\frac{u_{BE}}{U_T}} \end{array} \right.$$



$$u_O = -i_R R$$

$$\approx -I_S R e^{\frac{u_I}{U_T}}$$



Example: Use operational amplifier to realize the function below and draw the circuit. Requirement: use 3 Op Amps, and the value of component is listed below:

$$C = 1\mu\text{F} \quad 1\text{k}\Omega \leq R \leq 1\text{M}\Omega$$

$$u_o = 2u_{i1} + 3u_{i2} - \int u_{i3} dt$$

Answer:

$$u_o = -[-2u_{i1} - 3u_{i2} - (-\int u_{i3} dt)]$$

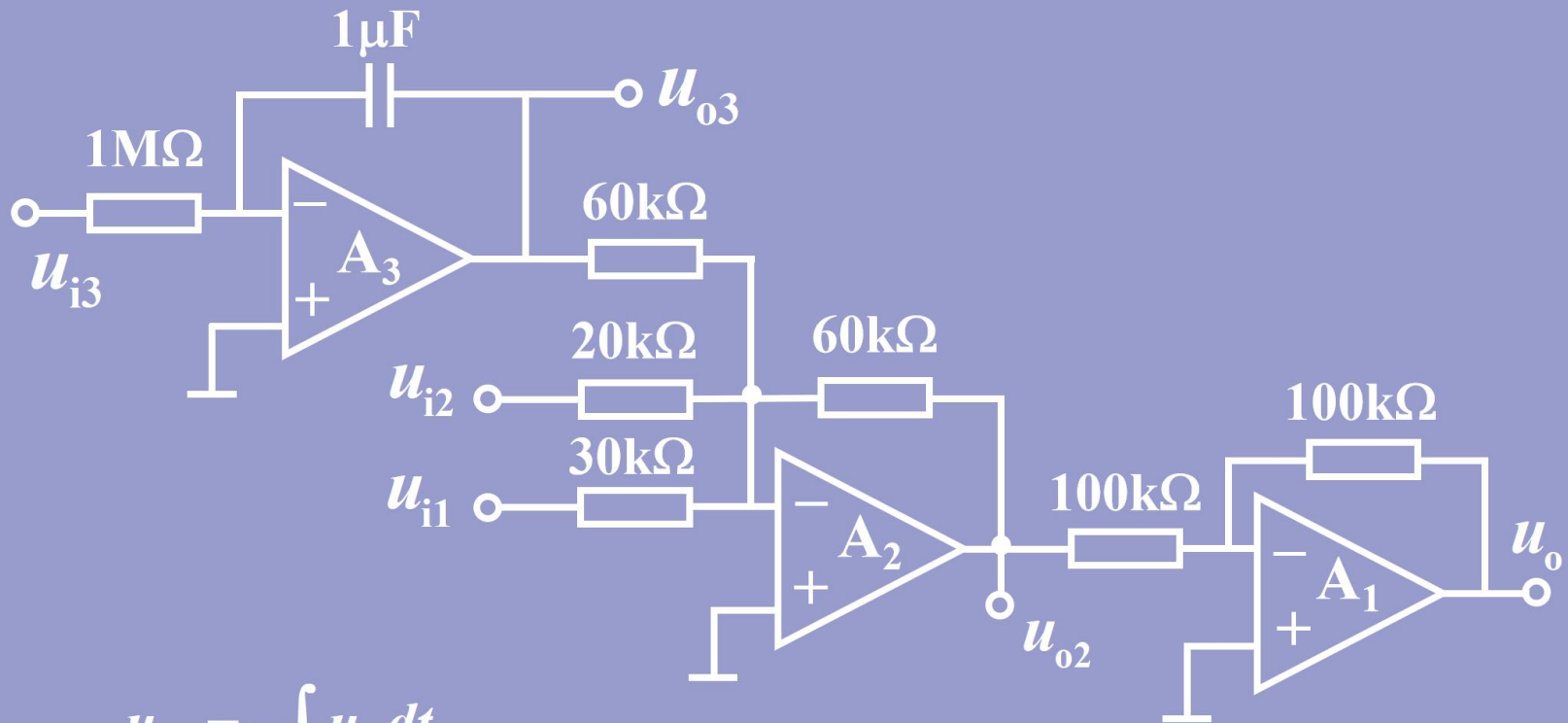
Then $-\int u_{i3} dt$ can be realized using integration operation

Equation $-2u_{i1} - 3u_{i2} - (-\int u_{i3} dt)$

can be realized using inverted addition operation

Another inverted scaling operation

$$u_o = -[-2u_{i1} - 3u_{i2} - (-\int u_{i3} dt)]$$

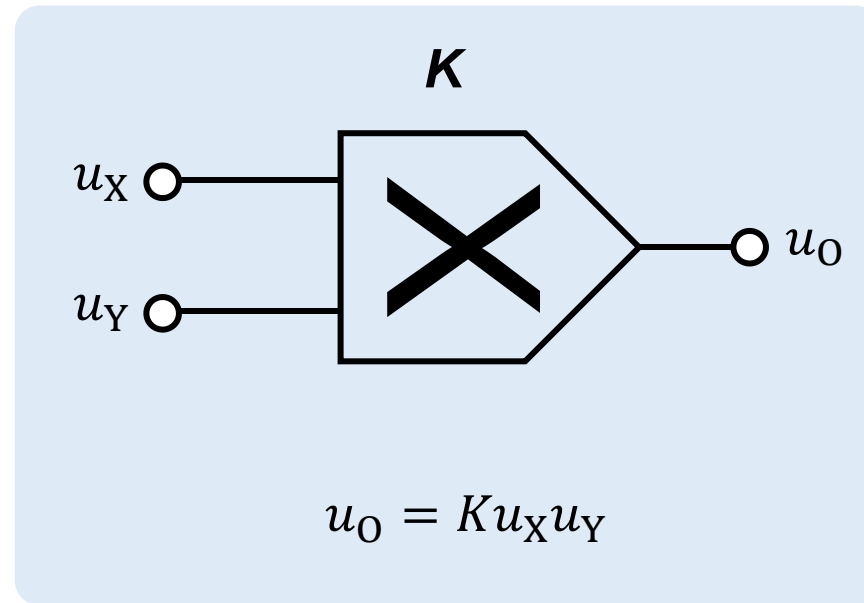


$$u_{o3} = -\int u_{i3} dt$$

$$u_{o2} = -2u_{i1} - 3u_{i2} - (-\int u_{i3} dt)$$

$$u_o = -[-2u_{i1} - 3u_{i2} - (-\int u_{i3} dt)]$$

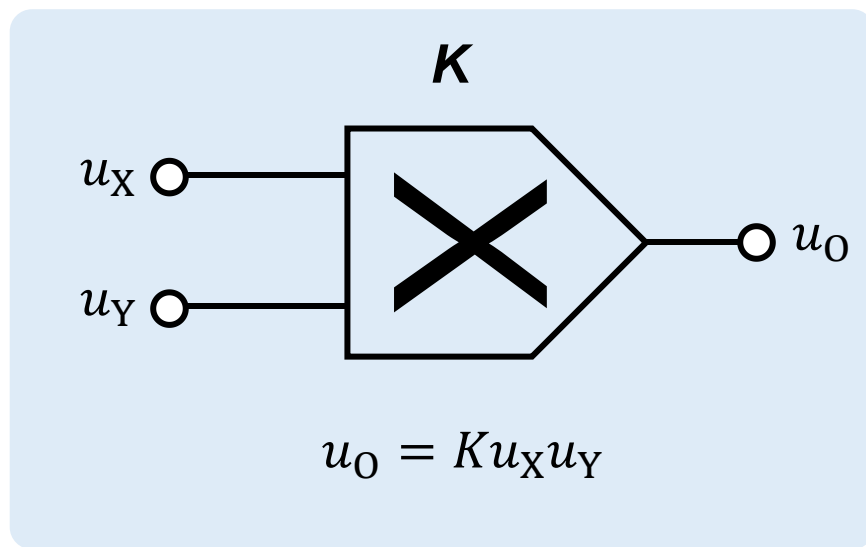
7.10 Analog multiplier 模拟乘法器



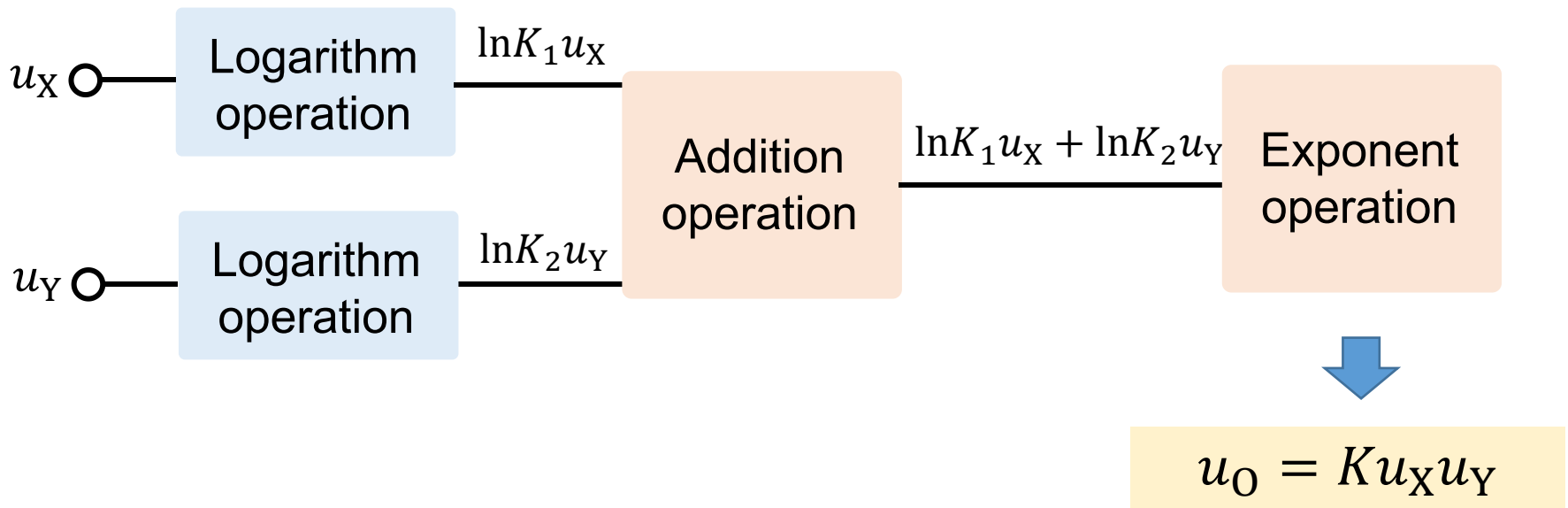
To realize multiplication function 实现乘法运算

(1) Multiplication operation using logarithms amplifier 对数乘法器

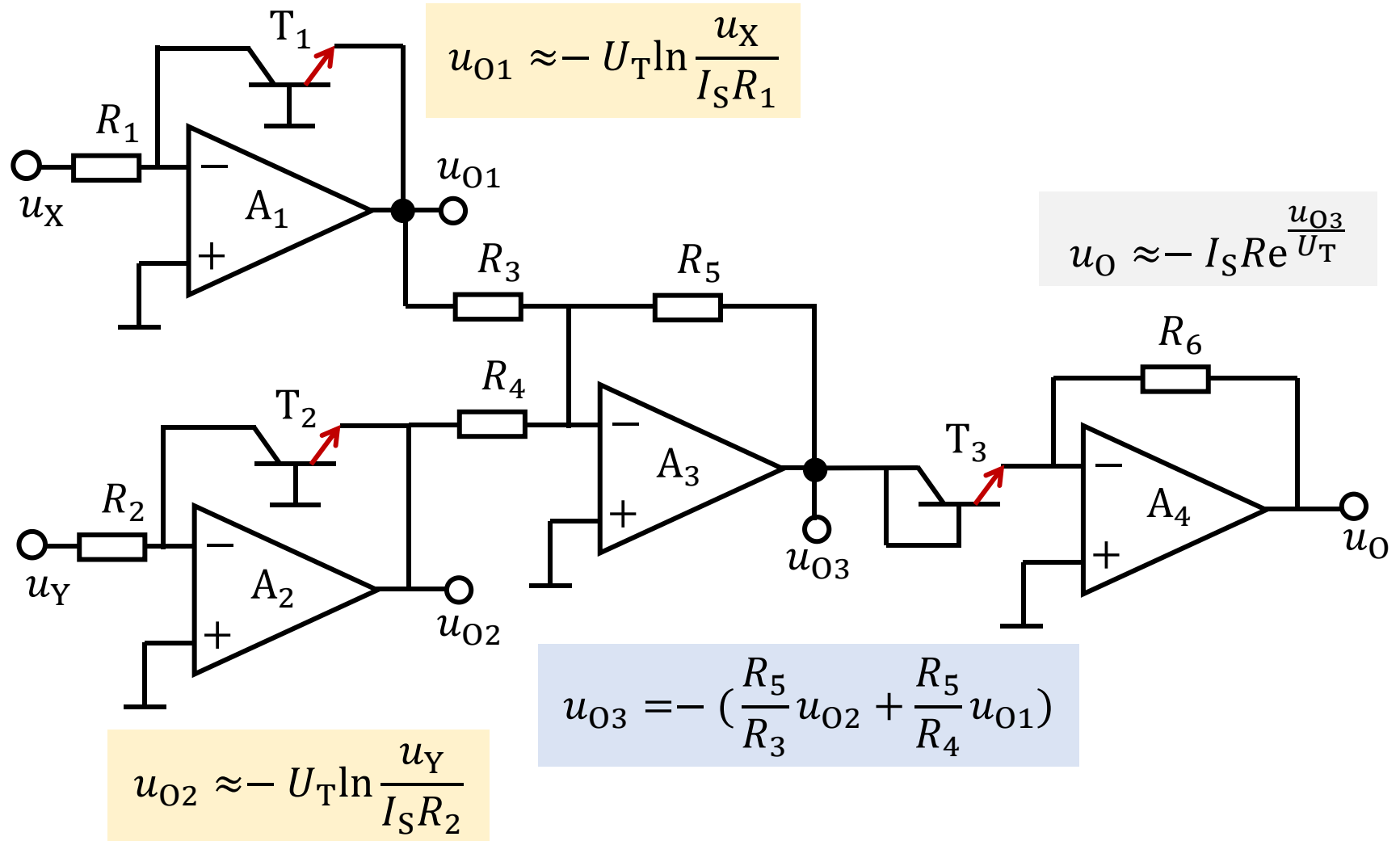
$$\begin{aligned}u_O &= Ku_X u_Y \\&= e^{\ln Ku_X u_Y} \\&= e^{\ln K_1 u_X + \ln K_2 u_Y} \\&, K = K_1 K_2\end{aligned}$$



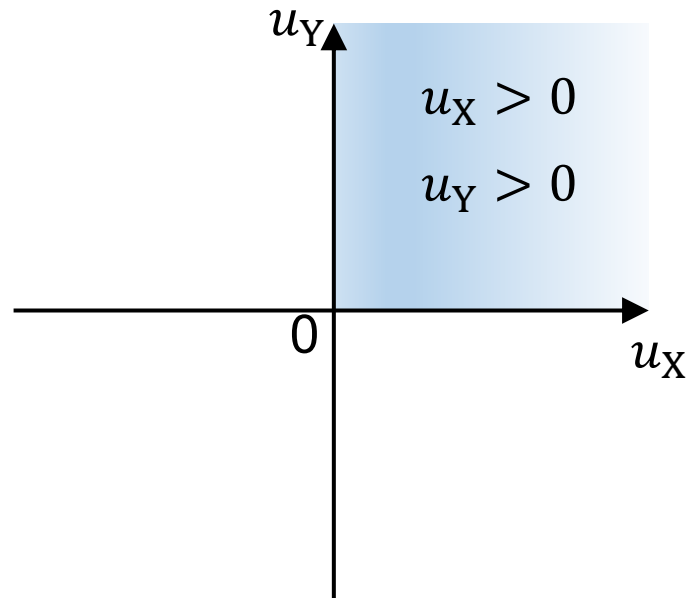
$$u_O = Ku_Xu_Y = e^{\ln Ku_Xu_Y} = e^{\ln K_1u_X + \ln K_2u_Y}, K = K_1K_2$$



Multiplication operation using logarithms amplifier



Multiplication operation using logarithms amplifier



(2) Variable transconductance multiplier 变跨导式乘法器

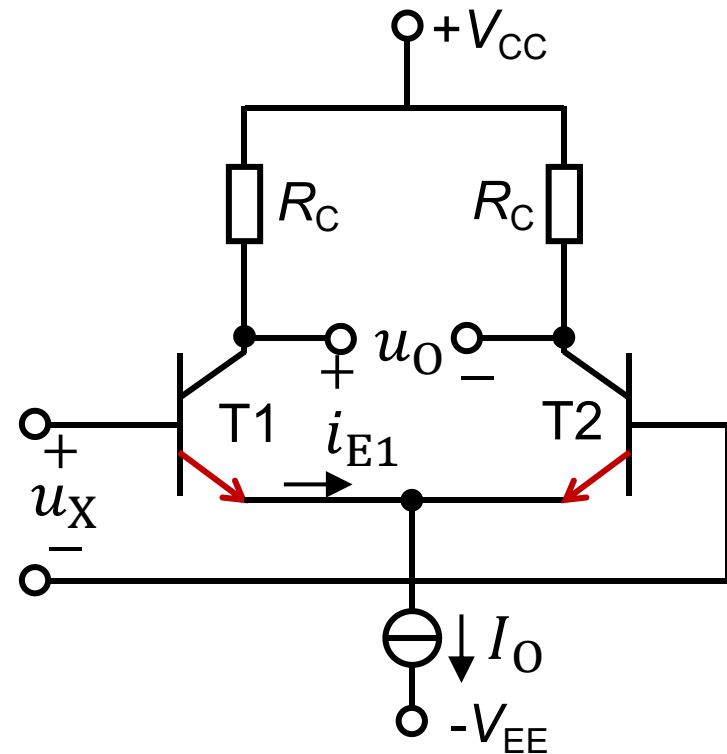
Working principle: using differential amplifier

$$\dot{A}_d = \frac{u_o}{u_X} = - \frac{\beta(R_C // R_{L/2})}{r_{be}}$$

$$g_m = \frac{\beta_0}{r_{be}} \approx \frac{I_{EQ1}}{U_T}$$

$$u_o = - \frac{\beta R_C}{r_{be}} u_X$$

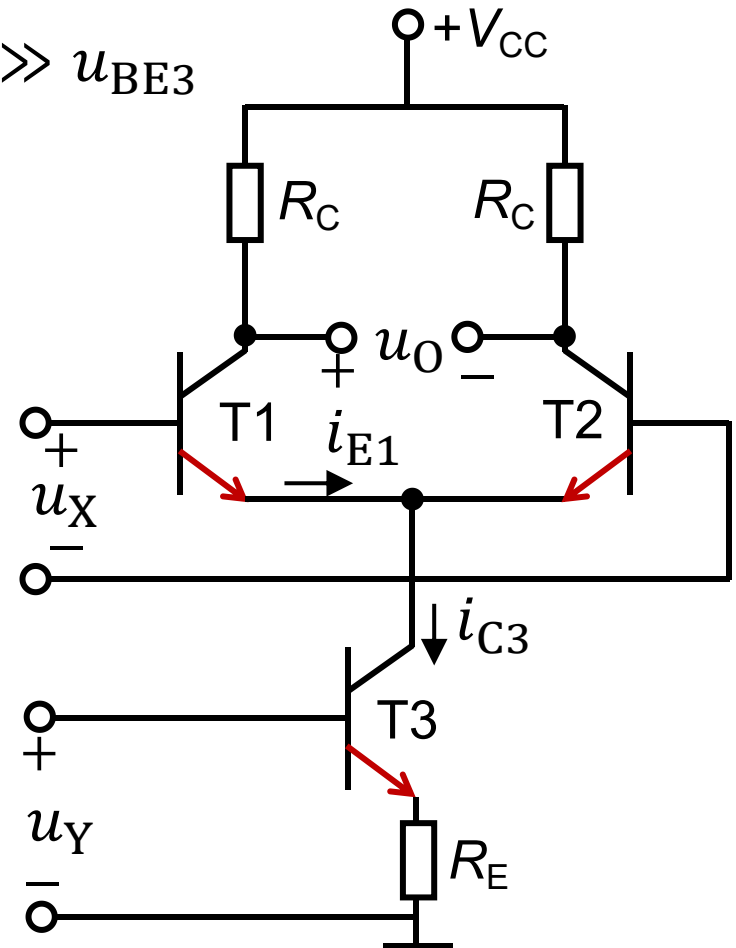
$$= - g_m R_C u_X$$



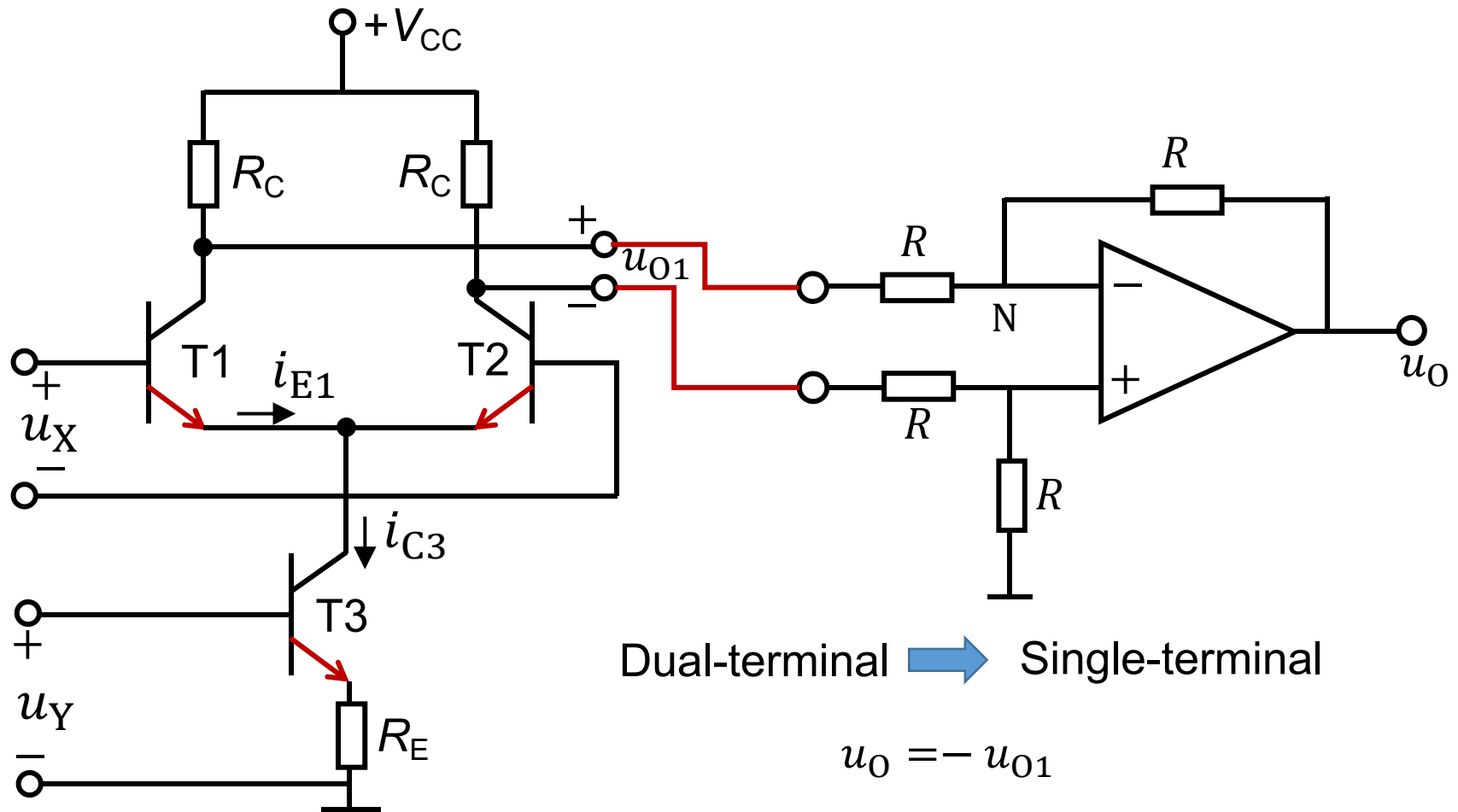
Differential amplifier with constant-current source as a multiplier

$$\left\{ \begin{array}{l} i_{C3} = \frac{u_Y - u_{BE3}}{R_E} \approx \frac{u_Y}{R_E}, u_Y \gg u_{BE3} \\ u_O = -g_m R_C u_X \end{array} \right.$$

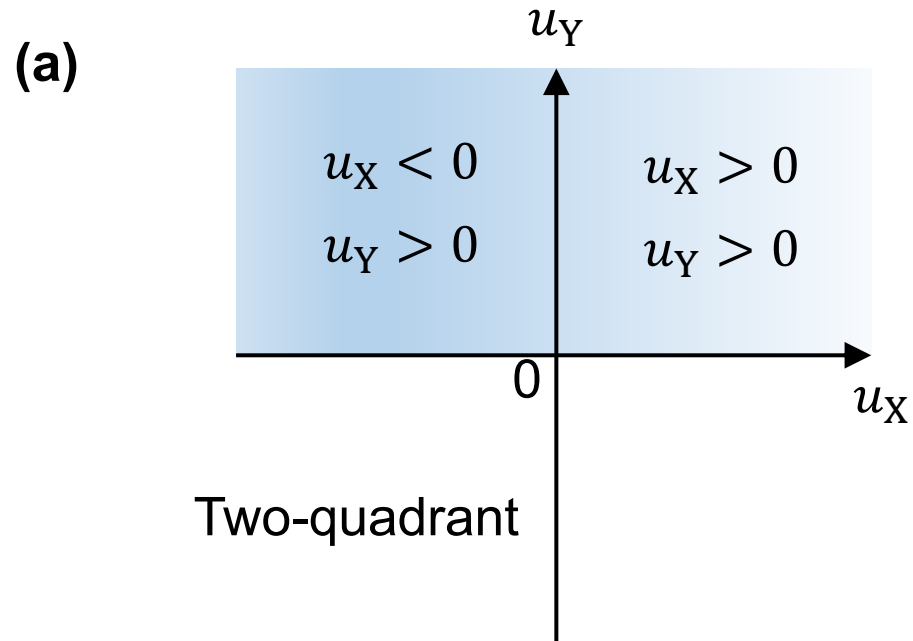
$$\begin{aligned} u_O &= -\frac{I_{EQ1}}{U_T} R_C u_X \\ &= -\frac{i_{C3}}{2U_T} R_C u_X \\ &\approx -\frac{R_C}{2U_T R_E} u_X u_Y \end{aligned}$$



Dual-terminal to single-terminal

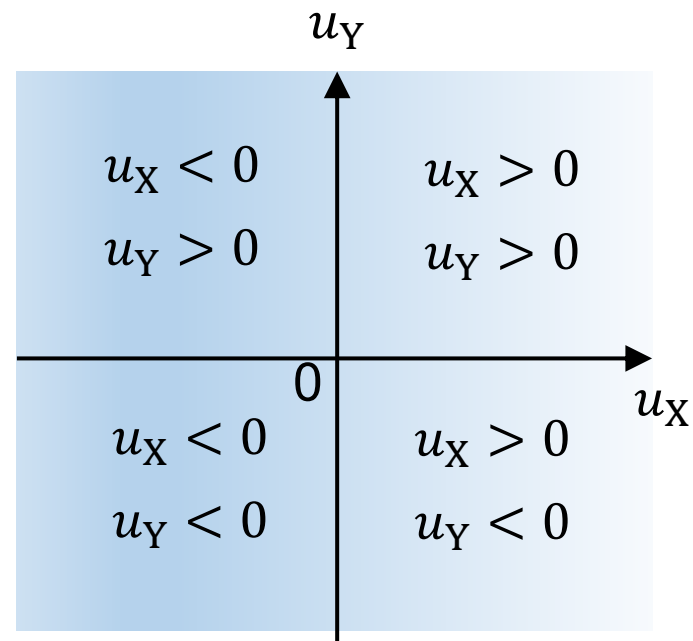
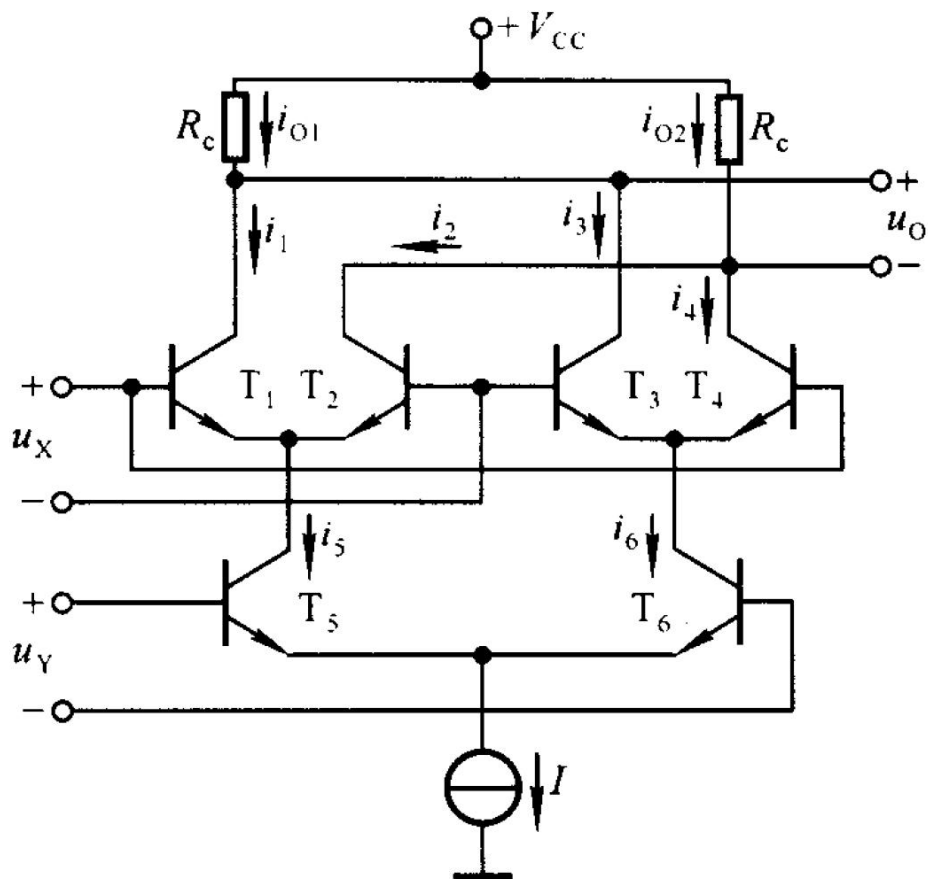


Drawback



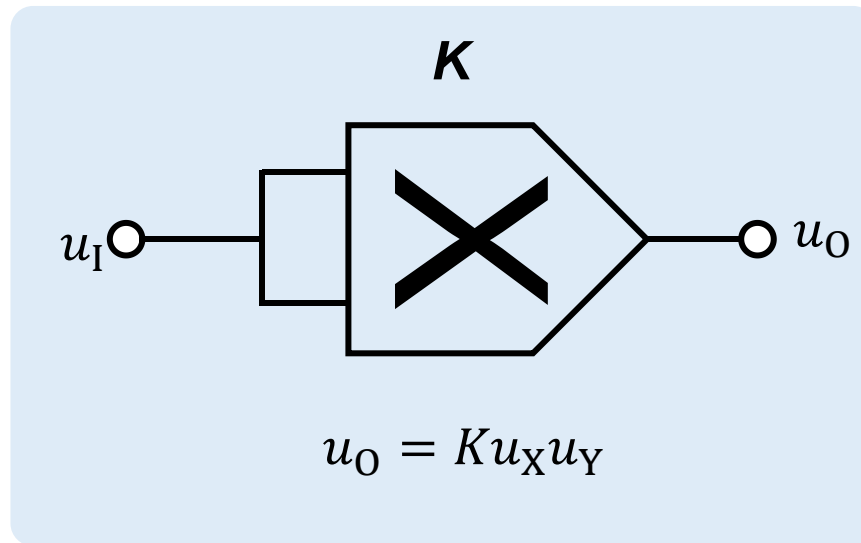
(b) Smaller u_Y , larger error

Four-quadrant 四象限:



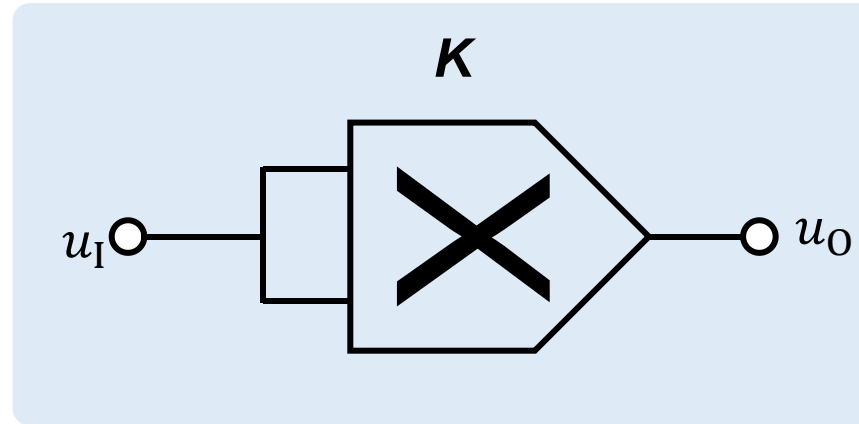
(3) Application of Multiplier

a) Square operation 平方运算



$$\begin{aligned} u_O &= Ku_X u_Y \\ &= Ku_I^2 \end{aligned}$$

a) Square operation 平方运算



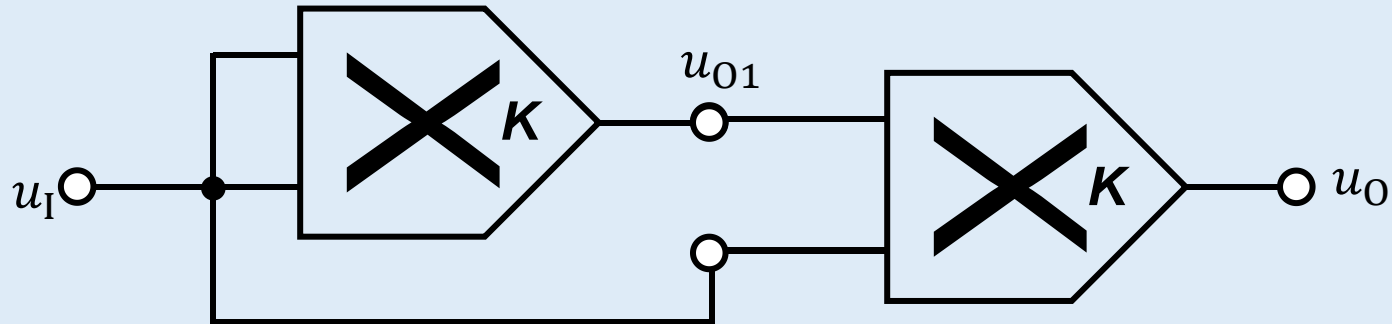
$$u_O = Ku_I^2$$

When $u_I = \sqrt{2}U_i \sin \omega t$,

$$u_O = 2KU_i^2 \sin^2 \omega t = KU_i^2 (1 - \cos 2\omega t)$$

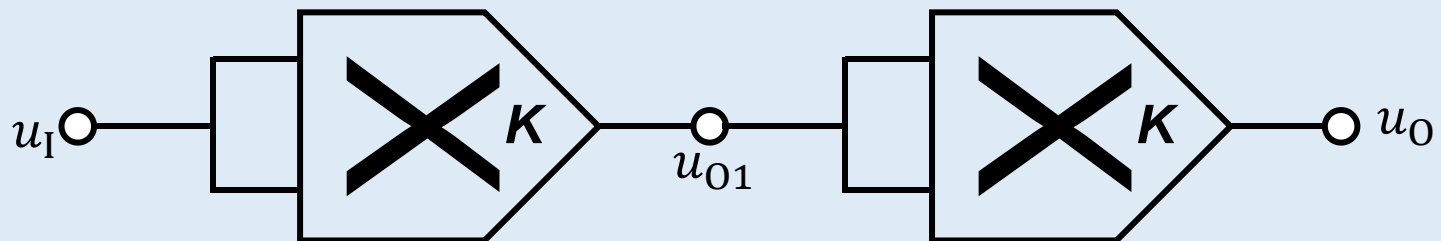
The output frequency is doubled!

b) Cubic operation 立方运算



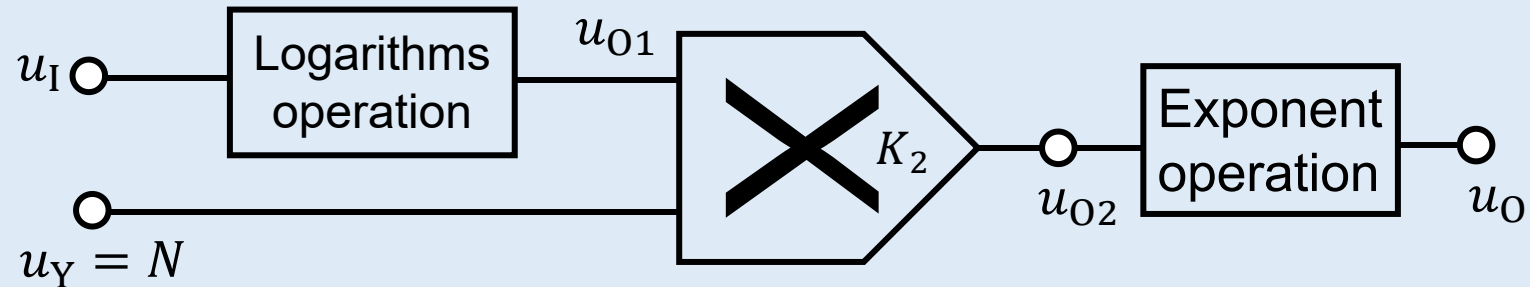
$$u_O = K u_{O1} u_I = K(K u_I^2) u_I = K^2 u_I^3$$

c) The 4th-power operation 四次方运算



$$u_O = K u_{O1}^2 = K(K u_I^2)^2 = K^3 u_I^4$$

d) How to realize n^{th} -power operation?



$$u_{O1} = K_1 \ln u_I$$

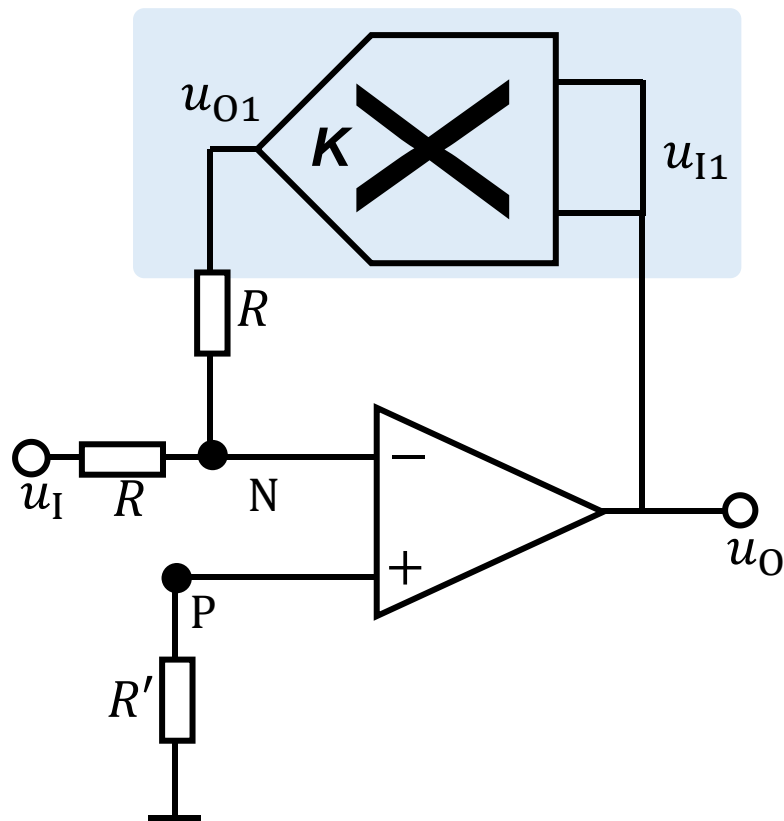
$$u_{O2} = K_1 K_2 N \ln u_I$$

$$u_O = K_3 \exp(K_1 K_2 N \ln u_I)$$

$$u_O = K_3 u_I^{K_1 K_2 N}$$

$$\text{Let } K_1 K_2 = 1, \quad u_O = K_3 u_I^N$$

f) **Square root (Sqrt) operation** 开平方运算

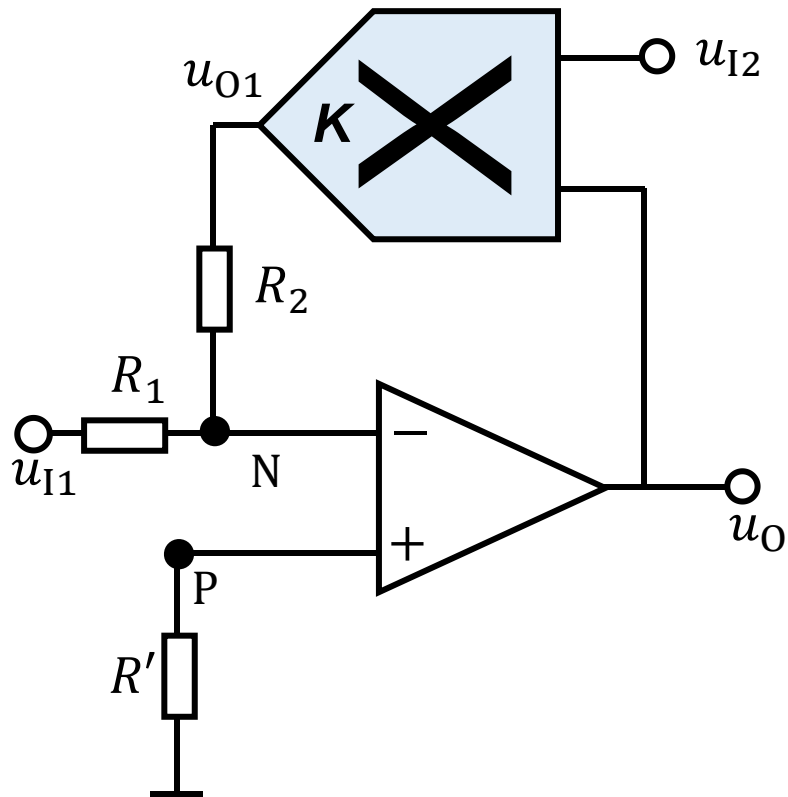


$$u_{O1} = Ku_O^2$$

$$\frac{u_{O1}}{R} = -\frac{u_I}{R}$$

$$u_O = \sqrt{-\frac{u_I}{K}}$$

g) Division operation 除法运算



$$u_{O1} = K u_O u_{I2}$$

$$\frac{u_{O1}}{R_2} = - \frac{u_{I1}}{R_1}$$

$$u_O = - \frac{R_2}{K R_1} \frac{u_{I1}}{u_{I2}}$$

Homework 8: Refer to blackboard

Deadline: 2025-05-28 11:59pm